

The Price of Thought: wiring cost and the brain–nerve cord connective in the complete *Drosophila* male CNS connectome

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Abstract

The complete connectome of the adult male *Drosophila* central nervous system (neuPrint male-cns:v1.0) was represented as a directed graph of 23,073 cell types resolved by hemisphere, with 490,884 connections and each type placed at the centroid of its neurons' cell bodies. Three questions were pre-registered.

First, placement. The real layout costs 0.459 times the mean of 1000 random reassignments of the same positions ($z = -278.8$, $p = 0.0010$; none as cheap), and 0.728 times when positions are shuffled only within brain or nerve cord. It is not a local optimum: cost-reducing swaps lower the total by at least 33.6%.

Second, the neck connective of descending and ascending neurons. Its neck-crossing connections are 7.5% of all connections but 24.0% of wiring cost. Their partners are enriched for high-degree types, and connective types are over-represented among hubs (odds ratio 3.22). Rich-to-rich routes through the connective exceed a degree-preserving null by 1.103 ($z = 4.8$, $p = 0.0010$). That excess vanishes at a lower input threshold, its split between ascending and descending routes reverses across thresholds, and a model using only distance, compartment and cell class reproduces it.

Third, value. Cutting the connective removes 0.36 times the sensory-to-motor flow capacity lost with random wiring of equal total length ($z = -42.5$, $p = 1.0$ against the registered direction), but 1.83 times in the brain-to-nerve-cord direction ($z = 48.2$, $p = 0.0010$). Across cell types, wire length does not predict flow once the number of connections is controlled for (partial $\rho = 0.044$ descending, -0.122 ascending).

Eight exploratory analyses accompany these tests. Attributed to the neuropils, only 6.6% of the wiring budget stays inside one region, and all 48 regions large enough to test are wired internally more cheaply than permutations of their own cell types, the antennal lobes least clearly. Over connections the budget is unevenly but not extremely spread: the longest tenth hold 30.7% of it, and a lognormal fits the upper tail better than a power law.

Five of them were run later still, and none had a direction fixed in advance. Degree does not predict how centrally a cell type sits ($\rho = -0.043$), but a type sits 127.3 μm from the centroid of its own partners against 320.7 μm for degree-matched random partners. The two hemispheres hold wire within a median factor of 1.13, with no side bias a sign-flip null can separate from chance. Connection length and synapse count are all but uncorrelated ($\rho = -0.054$), so the extra wire of a long connection buys reach and not synaptic weight. The 89 neuropils form a network that is dense, not small-world by wire ($\sigma = 0.109$), and split by Louvain into two communities that follow the brain and the nerve cord. Removing a quarter of the wire from the long end of the length distribution costs 6.5% of the graph's efficiency against 28.3% from the short end, although matched on connection count the longest connections are individually worth more.

The fly concentrates its most expensive wiring on well-connected cell types. That wiring carries no more sensory-to-motor flow per unit length than ordinary long wiring, except from brain to nerve cord. Four of the thirteen registered hypotheses failed, and both the rich-to-rich excess and the direction it comes from change with the input threshold, so these are statements about one static wiring diagram under one set of definitions.

Introduction

This report is the third part of a three-part study of one connectome, the complete male *Drosophila* central nervous system (Berg et al. 2026). ConnectomeLens asked whether the male wiring alone carries a detectable signature of the cell types annotated as sexually dimorphic. Fault Lines asked how much of the nervous system has to be removed before sensory input no longer reaches motor output. This part asks what the wiring costs in physical length, and what the most expensive part of it, the connective between brain and nerve cord, buys in return.

Wiring economy is the idea that nervous systems place their components so as to keep the total length of their connections small. Ramón y Cajal (1909–1911) stated it as a law of conservation of space, time and material. Cherniak (1994, 1995) treated it as a component-placement problem. He found the layout of *C. elegans* ganglia and of mammalian cortical areas at or near the cheapest of the arrangements he examined.

Later work qualified the result:

- Chen, Hall and Chklovskii (2006) found the *C. elegans* layout cheaper than random, yet well above the optimum they computed for it.
- Kaiser and Hilgetag (2006) showed that rearranging components can cut wiring by about 48%, because long-range projections that shorten processing paths are kept. Ahn, Jeong and Kim (2006) reached a similar conclusion.
- Gushchin and Tang (2015) found the real *C. elegans* layout 30.6% cheaper than random, while a constrained optimization of 86 interneurons cut a further 35.1%.

Placement cheaper than random is therefore not placement that is optimal, and the two claims are tested separately here.

Rich-club organization is the tendency of highly connected vertices to be more densely interconnected than their degrees alone predict (Colizza et al. 2006). In the human connectome the rich club is costly. Its connections are long and make up a disproportionate share of total wiring, and they carry a disproportionate share of the shortest communication paths (van den Heuvel and Sporns 2011; van den Heuvel et al. 2012). The same high-cost, high-capacity pattern has been reported in *C. elegans*: the connections of an 11-neuron rich club account for 48% of total wiring cost (Towlson et al. 2013).

The fly’s neck connective is the most obvious candidate for such a backbone. Descending neurons carry signals from brain to nerve cord and ascending neurons from nerve cord to brain. Each must span the neck, so their connections cannot share in the local economy that keeps most wiring short.

Both ideas have already been examined in the fly brain, and this study extends that work rather than starting it:

- Rivera-Alba et al. (2011) found the arrangement of neurons in a lamina cartridge cheaper than a million random rearrangements.
- Salova and Kovács (2025) shuffled neuron positions across the hemibrain and found shuffled layouts about twice as costly as the real one. They also found connection probability decaying exponentially with distance, and fitted generative models combining degree with wiring length.
- Péntek and Ercsey-Ravasz (2025) fitted an exponential distance rule model to FlyWire neuropils.
- Lin et al. (2024) found a neuron-level rich club in the FlyWire brain against a degree-preserving null. They expected ascending and descending neurons to belong to a rich club spanning the whole central nervous system, and noted that testing this required a complete CNS connectome.

Two recent network analyses of the FlyWire brain connectome (Dorkenwald et al. 2024; Schlegel et al. 2024) are the closest prior work:

- Grindrod, Lambiotte and Sahasrabuddhe (2024) studied 32,272 central-brain neurons. They found 67 Infomap modules, 21 of which carry 94% of the flow, arranged in a significant SpringRank hierarchy and spatially interleaved rather than compact.
- Sulyok, Balogh and Palla (2026) compared the real three-dimensional coordinates of 132,483 neurons with hyperbolic and Node2vec embeddings. Real coordinates predict connections with an AUC of 0.862 but support greedy routing in only 7.5% of attempts, and every embedding outperforms them.

Neither paper measures wiring cost, tests a rich club, fits a generative wiring model, or contains the nerve cord. In FlyWire, ascending and descending neurons appear only as their parts inside the brain; in the male ventral nerve cord connectome (Takemura et al. 2024), only as their parts inside the cord.

The two whole-CNS connectomes have already identified the connective as a bottleneck, by other measures. Berg et al. (2026) computed maximum flow from sensory modalities to motor domains and found descending and ascending neurons with much higher flow utilization than other classes. Bates et al. (2026), in a female brain-and-nerve-cord connectome, found them with high betweenness on sensory-to-effector paths. Neither measures what the connective costs in wire, whether it favours highly connected partners, or whether removing it does more damage than removing ordinary wiring of the same length.

To our knowledge, this is the first test of wiring economy and rich-club organization on a single synapse-resolution connectome of an entire central nervous system, with the brain–nerve cord connective treated as its own class of connections. It adds four things at cell-type resolution:

1. a placement permutation test spanning brain, nerve cord and connective;
2. a degree-preserving test of whether connective routes join high-degree types on the two sides, which addresses the expectation of Lin et al. (2024) directly;
3. the loss of sensory-to-motor flow capacity when the connective is removed, against removing non-connective wiring of equal total length;
4. a test of whether a generative model built on distance, compartment and cell class reproduces these statistics.

Data and Methods

Spatially embedded cell-type graph

All data were retrieved from neuPrint (Plaza et al. 2022), dataset `male-cns:v1.0`, with `neuprint-python`; coordinates were converted at the dataset’s 8 nm voxel size, and the somata of typed neurons span $730 \times 514 \times 995 \mu\text{m}$. The 164,506 typed neurons were grouped so that each vertex of the graph is one annotated cell type on one side of the body (the soma hemisphere, or the nerve-root hemisphere for neurons without a CNS soma), since pooling a bilateral type would place it on the midline. This gives 23,073 vertices and, at the threshold set below, 490,884 directed arcs.

Below, a *cell type* always means one of those vertices and a *connection* one of those arcs. The words node and edge are kept only inside the standard technical terms that have no other name: in-degree and out-degree, edge-disjoint paths, degree-preserving edge swaps, edge betweenness, and the non-edges sampled by the generative model.

A cell type sits at the centroid of its neurons’ soma positions when at least half of them have one (22,139 types). Otherwise it sits at the synapse-weighted centroid of its neurons’ synapses (934 types, 751 of them sensory, whose cell bodies lie in the periphery). A cell type is brain-dominant if more than half of its synapses in CentralBrain, Optic(L), Optic(R) and VNC lie in the first three, and nerve cord-dominant otherwise: 16,052 and 7,021 types.

A directed connection joins two cell types when it supplies at least 1% of the target’s input synapses from typed neurons; self-connections are excluded. The same rule applied to types pooled across sides reproduces exactly the 11,751-type, 243,439-connection graph used in ConnectomeLens and Fault Lines.

The connective

Each cell type takes the superclass held by most of its neurons. Descending types (`descending_neuron`) number 480 pooled across sides and ascending types (`ascending_neuron`) 563; resolved by hemisphere they form the connective set of 951 descending and 1,096 ascending cell types, 8.9% of all cell types. As a cross-check, 85.4% of typed neurons with synapses inside the neck connective ROI (CV) are descending or ascending neurons. Sensory ascending neurons, whose axons also pass through the neck, have no CNS soma to place and are treated as sensory, not connective.

Partners are all non-connective cell types, split by compartment into brain partners B (14,950) and nerve-cord partners V (6,076). A neck-crossing connection joins a connective cell type to a partner on the far side of the neck: $\text{DN} \rightarrow \text{V}$, $\text{V} \rightarrow \text{DN}$, $\text{AN} \rightarrow \text{B}$ or $\text{B} \rightarrow \text{AN}$. There are 36,943 such connections.

Wiring cost

The cost of a connection is the Euclidean distance between its endpoints’ positions; the cost of a set of connections is the sum over the set, each connection counted once. Connection lengths have median $154.3 \mu\text{m}$ and mean $189.6 \mu\text{m}$. Throughout, the length of a single connection is given in micrometres and the summed length of a set of connections in millimetres.

Cell-type positions stand in for the length of real neuronal processes. As a check, skeletons were retrieved for a stratified random sample of 500 neurons: 100 descending, 100 ascending, 150 central-brain intrinsic and 150 nerve-cord intrinsic. Their total cable length was compared with the distance from soma to the centroid of the neuron’s presynaptic sites.

Placement test

The primary test compared the real total unweighted cost with 1000 uniform random permutations of the position-to-type assignment, with the graph and the multiset of positions unchanged. Secondary versions were:

- the synapse-weighted cost;
- permutation only within compartments;
- synapse centroids as positions for every cell type;
- the brain and nerve-cord subgraphs separately.

Distance from a local optimum was estimated with 2,000,000 proposed swaps of position between two soma-placed cell types of the same compartment, each accepted if it lowered the total cost, following Kaiser and Hilgetag (2006) and Gushchin and Tang (2015). Connection probability was measured in $20 \mu\text{m}$ distance bins, and an exponential $P(d) = a \cdot \exp(-d/\lambda)$ was fitted over $0\text{--}600 \mu\text{m}$, over all pairs and separately within the brain, within the nerve cord and across the two compartments.

Wire by neuropil

The 105 published neuropil surfaces were simplified onto a $3 \mu\text{m}$ grid and pooled into one search tree. Each cell type takes the neuropil whose surface lies nearest its position, and each connection lends half its length to the neuropil at

each of its ends, so the wire attributed to the regions sums to the total. A neuropil's internal connections are those with both endpoints assigned to it.

Placement inside a region was tested with the statistic of the primary placement test restricted to that region: the summed length of its internal connections against 1000 permutations of which of its own cell types sits at which of its own positions. It is reported for neuropils with at least 12 cell types and 50 internal connections. These region tests were not planned as a family and carry no registered hypothesis; where they are read as significance, a Bonferroni threshold across the 48 tested regions is stated alongside.

Concentration of the wiring budget

Connections were sorted longest first and the cumulative share of wire taken against the cumulative share of connections; the Gini coefficient is one minus twice the area under the same curve taken shortest first. It was computed over all connections and over those within the brain, within the nerve cord and across the neck.

The upper tail of the length distribution was fitted with the `powerlaw` package (Alstott et al. 2014), which selects the lower bound minimizing the Kolmogorov-Smirnov distance above it (Clauset et al. 2009) and then fits the exponent by maximum likelihood. Lengths were rounded to the micrometer first, which cuts the candidate lower bounds from half a million to about a thousand without moving the fit by more than the rounding. The fit was weighed against a lognormal, an exponential and a truncated power law by the normalized log-likelihood ratio of Vuong (1989), a positive ratio favouring the power law.

Wire was also attributed to cell classes, each connection counted for the majority superclass at each of its two ends. That majority is taken within a cell type on one side, and differs a little from the connective set above, which takes it over the cell type pooled across sides: 1,089 types have `ascending_neuron` as their majority superclass, against the 1,096 ascending types of the connective set.

Rich-club tests

A partner's richness is its total degree in the subgraph of non-connective cell types, so no connective connection contributes to it. Rich brain and rich nerve-cord partners are those at or above the 90th percentile of richness within B and within V. A route is a two-connection path through one connective cell type, $b \rightarrow \text{DN} \rightarrow v$ or $v \rightarrow \text{AN} \rightarrow b$. The statistic R is the number of rich-to-rich routes: the sum over connective cell types of their rich inputs times their rich outputs on the opposite side.

The four layers of connective connections ($B \rightarrow \text{DN}$, $\text{DN} \rightarrow V$, $V \rightarrow \text{AN}$, $\text{AN} \rightarrow B$) were randomized independently by degree-preserving edge swaps (Maslov and Sneppen 2002; 10 swaps per connection, no multi-edges), 1000 times. Every partner keeps its number of connective connections in each layer and every connective cell type keeps its in- and out-degree; only which partner attaches to which connective cell type changes.

The registered richness threshold is the top 10%. R was also computed at the top 1, 2, 5, 20, 30 and 50%, as an exploratory curve around it; the treatment of those six is set out under Robustness and multiple comparisons.

Two secondary tests were run:

- whether rich partners are over-represented among connective partners, against a null that redraws each connective cell type's partners uniformly;
- whether descending and ascending types are over-represented among cell types at or above the 90th percentile of total degree (Fisher exact test).

The rich-club coefficient of the whole graph was normalized by 100 degree-preserving randomizations.

Value tests

Sensory-to-motor value uses the definitions of Fault Lines, with the code copied unchanged:

- **Flow capacity** is the maximum number of edge-disjoint directed paths from any sensory to any motor cell type, computed as a unit-capacity maximum flow between a supersource and a supersink; by Menger's theorem and max-flow min-cut duality (Menger 1927; Ford and Fulkerson 1956) the two quantities are the same.
- **Reachable pairs** is the number of (sensory, motor) pairs joined by a directed path.

The sensory set has 751 cell types (390 brain, 361 nerve cord) and the motor set 367 (86 brain, 281 nerve cord). Descending neurons, which Fault Lines counted among motor outputs, are excluded here because they are part of what is removed, so the intact capacities of the two studies are not comparable.

The primary test removed the 36,943 neck-crossing connections. It compared the loss with removals of 1000 random sets of non-connective connections matched to their total length: connections were taken in random order until the running total reached that length, and every set came within 0.01% of it. Secondary comparisons were:

- sets matched on connection count instead of length;
- the longest non-connective connections up to the same total length;
- silencing every connection incident on a connective cell type;
- flow in each direction across the neck.

For each connective cell type, *price* is the summed length of its neck-crossing connections. *Value* is the loss of flow in the direction the type carries when those connections alone are removed. Price and value were correlated across descending and across ascending types, then again with the number of neck-crossing connections partialled out.

Generative model

Connection presence over ordered pairs of cell types was modelled by logistic regression, fitted in scikit-learn (Pedregosa et al. 2011) on all 490,884 connections and 2,454,420 uniformly sampled non-edges. The intercept was corrected for the sampling ratio. Model G uses:

- distance, and the logarithm of distance;
- whether the two cell types share a compartment;
- a one-hot term for the ordered pair of superclasses.

A secondary model, G+deg, adds the logarithms of source out-degree and target in-degree. Fit is reported as McFadden pseudo- R^2 , AIC and 5-fold cross-validated ROC AUC.

Fifty synthetic graphs per model were drawn as independent Bernoulli trials, with a scalar shift solved so that the expected connection count equals the real one. Each synthetic graph was compared with the real graph on thirteen properties. A property counts as reproduced when the real value lies within the central 95% of the synthetic values.

Hub placement

Two statistics over the same graph. The first is the rank correlation between a cell type's total degree and its distance from the centroid of the cell types of its scope, over all types, over brain types and over nerve-cord types. Each scope takes its own centroid: the brain and the nerve cord occupy different volumes, and a shared centre would rank a type by which compartment it belongs to rather than by where it sits inside it. The same correlation was taken for the summed length of the connections a type takes part in, and Pearson's r between distance and the base-10 logarithm of each measure over the types with a positive value.

The second compares the distance from a cell type to the centroid of its partners with a null in which the type keeps its degree and draws that many partners uniformly at random from the other 23,072 cell types, with replacement and never itself; 1000 draws, seeded from 20360916. The three cell types with no connection are not tested. A scope's mean is compared with the mean of the same draws restricted to that scope.

Wire on the two sides

A neuropil name ending in (L) or (R) was split into a stem and a side, and stems present on both sides form a pair. Each side's wire, share, cell types and internal cost ratio are read from `results/wire_atlas.json` rather than recomputed. A pair's log ratio is the natural logarithm of the left side's wire over the right side's, so it is zero for a pair in balance.

Three nulls, seeded from 20410916, 20410917 and 20410918:

- if the two hemispheres are wired alike, the sign of a pair's log ratio is arbitrary, so negating each of the 36 ratios at random gives the null distribution of their mean; 1000 sign flips, two-sided;
- matching each left neuropil to a right neuropil drawn at random instead of to its own counterpart gives the asymmetry expected between two unrelated structures of this atlas, against which the observed mean absolute log ratio is one-sided over 1000 matchings. This null sets no scale for how alike two copies of one structure ought to be, so it can only rule out the case where the pairing carries no information at all;
- the same sign flip applied to the left-minus-right difference in internal cost ratio, over the pairs whose within-neuropil placement test ran on both sides.

Neuropils carrying no hemisphere suffix, and sides whose counterpart holds no cell type in this reconstruction, take no part in the three tests and are reported by name.

Synapses against length

The weight of a connection is the number of synapses it carries between the typed neurons of its two cell types, the same weight the input threshold is computed from. Connections were sorted by length into ten equal groups, and each group's synapses per micrometre taken as its summed synapse count over its summed wire, against the density it would show if synapse count did not depend on length at all: the mean synapse count over all 490,884 connections divided by the group's mean length. Length and synapse count were correlated by Spearman's ρ and by Pearson's r on the base-10 logarithm of both, over all connections, over the 36,943 that cross the neck and over the 453,941 that do not. Nothing here is randomized and no null is drawn.

The neuropil network

Cell types take the nearest of the 105 published neuropil surfaces and each connection lends half its length to the neuropil at each of its ends, by the rule of the wire atlas above. A connection with both ends in one region is that region's internal wire; the rest join two regions, and the summed length over a pair of regions is the wire of one undirected edge between them. The whole pair matrix is recomputed here because the atlas keeps only its 40 largest pairs, and the recomputation reproduces the atlas total and every pair it keeps.

Clustering is the mean local clustering coefficient, a region with fewer than two partners contributing zero, and path length the mean shortest path over all pairs of regions. The weighted versions weigh a pair by its wire, a step along the heaviest pair of regions costing one and a thinner pair proportionally more. Two nulls of 1000 draws each, both preserving every region's number of partners exactly:

- **rewired:** degree-preserving edge swaps randomize which pairs of regions are joined, at 10 swaps per edge and no multi-edges, and the multiset of per-pair wire values is then dealt out again over the rewired edges, so region strength is not preserved;
- **weights shuffled:** the real topology is left alone and only the wire values are permuted over its edges, so the unweighted measures are the real ones by construction and the comparison isolates the arrangement of wire over a fixed structure.

Null i of either kind is seeded from 20860916 + i , and p-values are two-sided, twice $(1 + \text{the smaller tail count}) / (1 + 1000)$. Communities come from Louvain on the wire-weighted region graph under 100 seeds from 20910916, the partition reported being the one returned most often. Its agreement with compartment, with hemisphere and with thoracic segment is the adjusted Rand index and the normalized mutual information over the regions carrying that label.

Removal from each end of the length distribution

Connections were removed cumulatively under three schedules — longest first, shortest first, and at random — and the graph left behind measured at 18 points placed at shares of the total wire from 0 to 0.6. At every point the longest-first and shortest-first schedules have removed the same amount of wire; the random schedule has removed the same *number* of connections as the longest-first one and is the mean of five draws, draw i seeded from 21010916 + i .

Two measures. The first is sampled global efficiency: the mean of $1/d$ over ordered pairs of cell types, d the directed shortest path in connections, estimated on the rows of the distance matrix belonging to 300 source cell types drawn once without replacement from 20960916 and used at every point of every schedule, so the curves differ only in what was removed. Characteristic path length is the more familiar measure and the wrong one here, because removal is exactly what makes pairs unreachable and the mean of a set containing infinities is undefined, whereas an unreachable pair contributes a well-defined zero to efficiency. The matrix is $23,073 \times 23,072$ pairs at every point of every schedule, which is why it is sampled rather than computed whole. The second measure is the number of cell types in the largest weakly connected component.

Robustness and multiple comparisons

The placement, route and hub tests were repeated at input thresholds of 0.5%, 2% and 5%, with positions and compartments unchanged. Empirical one-sided p-values are $(1 + \text{null values at least as extreme}) / (1 + N)$, so the smallest attainable with 1000 draws is 0.0010.

Two families of tests are larger than the hypotheses that govern them. The rich-to-rich route statistic was computed at seven richness thresholds, of which only the top 10% is pre-registered; the other six are exploratory. Their p-values are reported uncorrected and beside a Bonferroni threshold across the seven, $\alpha = 0.05/7 = 0.0071$. The within-neuropil placement tests cover 48 regions and carry no registered hypothesis; they are likewise reported uncorrected and beside $\alpha = 0.05/48 = 0.00104$.

The five analyses added after the registered tests do not make a third family of that kind. Each computes a different statistic on a different object, rather than one statistic over a sweep of a nuisance parameter or over a list of regions; none was selected for reporting on its p-value, and every statistic computed is reported whether or not it reaches one. Two of the five draw nulls. The three partner-distance comparisons of the hub-placement analysis all sit at the smallest one-sided value 1000 draws allow, 0.0010. Of the eight region-network comparisons, the six that are not fixed by construction sit at the smallest two-sided value, 0.0020, which is below a Bonferroni threshold across the eight, $\alpha = 0.05/8 = 0.0063$; the other two are reproduced exactly by the weight-shuffled null and return $p = 1$. The three side-symmetry nulls are reported individually and two of them return nothing. No other analysis in this report involves a family of tests.

Pre-registration

Thirteen hypotheses were registered across three commits. Each was written into the hypothesis section of its results file and committed before the statistic it governs was computed: 39892c5 (placement, before the spatial graph existed on disk), e62c6df (rich club, value, the wiring-economy extensions, the cable-length check, and the generative-model protocol) and 73fba33 (price against value, input-threshold robustness). The scripts only append beneath those sections. `verify/check_preregistration.py` checks that every hypothesis section is unchanged since its registration commit, that each commit predates the first commit of the results it governs, and that each results JSON records the same commit; `results/preregistration.md` indexes them and prints the command that recovers the wording at each commit. Four of the thirteen failed: H4, H10, H10b and H11 (Table 1).

The wire atlas, the concentration of the wiring budget and the within-neuropil placement tests carry no registered hypothesis. All three were run after the registered tests and are reported as exploratory description.

Five further analyses — hub placement, the wire on the two sides, synapses against length, the neuropil network, and removal from each end of the length distribution — were run later still, after all thirteen registered tests had been computed. None had a direction fixed in advance and none carries a hypothesis. They add nothing to the thirteen and change none of the four failures.

Software

Software versions are listed in Table 2 and every dependency is pinned in `requirements.txt`. The two typefaces the figures are set in, Archivo and Spline Sans Mono, are committed under `assets/fonts/` with their SIL Open Font Licenses, so a figure redraws to the same bytes rather than to whichever cut a font service happens to serve that day. The pipeline is a set of Python modules in `pipeline/`, run through `make` (Table 3).

`make reproduce` runs `data`, `analyze`, `hero`, `readme`, `paper`, `test` and `verify` in that order and rebuilds every result from `neuPrint`.

Randomness. The base seed is 20260916, and each analysis adds a fixed offset to it. The 1000 placement permutations are seeded from 20560916 and the greedy swap search from 21060916; the within-compartment placement permutations use 21110916 plus the compartment index. Layer randomization i of the rich-club test uses $20660916 + 10i$, the partner-redraw null of the endpoint-enrichment test 20660923 , and whole-graph randomization i $20710916 + i$. The value nulls use 20760916 plus a per-family and per-draw offset, the generative model's negative sampling 20860916, synthetic graph i of a model 20960916 plus a per-model and per-draw offset, and the cable-length sample 21160916. Within-neuropil permutation i uses $20260916 + i$. The input-threshold repeats reuse the seeds of the analyses they repeat. Of the five analyses added afterwards, the partner-redraw null of the hub-placement analysis uses 20360916; the side-symmetry nulls use 20410916 for the sign flip of the pairs' wire, 20410917 for the random re-matching and 20410918 for the sign flip of their internal cost ratios; region-network null i uses $20860916 + i$ and Louvain seed i $20910916 + i$; the 300 sources of the removal schedules are drawn once from 20960916 and random draw i of those schedules uses $21010916 + i$; synapses against length draws nothing. Two offsets are used twice: the region-network nulls share 20860916 with the generative model's negative sampling, and the removal-schedule sources share 20960916 with the first synthetic graph of model G. In each case the two draw different objects from separate generators, and neither analysis takes any input from the other.

Verification. `make verify` runs twenty-four check scripts in `verify/`, one for each result plus `check_preregistration.py`, `check_site_data.py`, `check_references.py`, `check_citations.py` and `check_document_numbers.py`. They check each result file against its own invariants and against the others, that the pre-registration sections are unchanged since their registration commits, that the data the site ships is a fresh assembly of those result files, that every file path named in the README, this report and the result files resolves, that every reference here is one verified in `results/prior_art.md` and is cited in the text, and that every headline number in this report and the README is the number in the result file it comes from. Checks whose inputs are absent report SKIP rather than

Table 1: The thirteen registered hypotheses and their outcomes, as indexed in `results/preregistration.md`. The generative model was registered as a protocol without a hypothesis: which of thirteen properties the model reproduces was to be reported either way.

	hypothesis	direction	outcome
H	real placement is cheaper than permutations of the same positions	lower cost	supported
H1	rich-to-rich routes through the connective exceed layer-preserving rewirings	more routes	supported
H2	rich partners are over-represented among connective partners	enriched	supported
H3	descending and ascending types are over-represented among hubs	odds ratio above 1	supported
H4	cutting the neck costs more flow than random wiring of equal length	more flow lost	not supported
H5	connection probability falls with distance	negative correlation	supported over all pairs
H6	the real placement is not a local optimum under swaps	saving above 1%	supported
H7	connections of high-degree cell types are longer	longer	supported
H8	connection length correlates with edge betweenness	positive	supported, weak
H9	skeleton cable length correlates with soma-to-output distance	positive	supported
H10	a connective cell type's wire length correlates with the flow it carries	positive, both directions	not supported
H10b	the correlation survives control for the number of connections	positive, both directions	not supported
H11	placement, routes and hubs keep direction and significance at 0.5%, 2% and 5%	unchanged	not supported

failing. Twenty-eight test modules in `tests/` cover graph construction, wiring cost, the placement permutation, the connective set, the rich-club and value statistics, the price analysis, rewiring, the connectivity metrics, the generative model, the input-threshold repeats, the wire atlas and wire concentration, the wiring-economy extensions, the cable-length check, hub placement, side symmetry, synapses against length, the neuropil network, the removal schedules, the renderings and the site export. `pytest` and `make verify` run in continuous integration on every push, alongside a build of the site. Every number reported here is read from a file under `results/`.

Availability. Code, derived results, figures and site data are available at <https://github.com/dhruvin-sarkar/price-of-thought> under the MIT license, with citation metadata in `CITATION.cff`; the raw neuPrint cache is not redistributed and is rebuilt by `make data`. An interactive presentation of the results is available at <https://dhruvin-sarkar.github.io/price-of-thought/>.

Table 2: Software versions.

package	version
Python	3.12.12
neuprint-python	0.6.3
igraph	1.0.0
NumPy / SciPy / pandas	2.5.3 / 1.18.1 / 3.0.5
scikit-learn	1.9.1
navis	1.12.0
powerlaw	2.0.0
networkx	3.6.1
matplotlib	3.11.2
pyarrow	25.0.1
pandoc / typst	3.11 / 0.15.1

Results

Placement is far cheaper than random, and far from optimal

The real placement costs 93,095 mm of wire, 0.459 times the mean of 1000 random reassignments of the same positions ($z = -278.8$). No permutation came as low ($p = 0.0010$, the smallest attainable). Every secondary analysis points the same way (Table 4, Figure 1):

The full permutation moves brain types into the nerve cord and is an easy null to beat. Shuffling only within compartments still leaves the real layout 27% cheaper, so the economy is not just the separation of brain and nerve cord.

Connection probability falls with distance (Spearman $\rho = -0.993$ across 51 bins with at least 1000 pairs), with a length constant of 76 μm over the first 600 μm . Within the brain and within the nerve cord the fall is not monotonic, and neither rank correlation is significant: $\rho = -0.154$ between brain cell types and $\rho = -0.159$ between nerve-cord cell types, $p = 0.14$ in both cases. Probability reaches a minimum at 340 μm between brain cell types and at 500 μm between nerve-cord cell types. It then rises again at the longest distances, where most connections join opposite sides of the body. The

Table 3: Make targets.

target	what it runs
make data	schema discovery, the spatially embedded cell-type graph
make analyze	placement permutations, rich club, value, wiring-economy extensions cable length, price against value, input-threshold robustness generative model and comparison, wire atlas, wire concentration hub placement, wire on the two sides, synapses against length the neuropil network, removal from each end of the length distribution
make hero	the hero rendering and the front view of the connective
make readme	README plates and figures, drawn from results/
make export	data for the interactive site
make paper	this report, with pandoc and the typst engine
make poster	the one-page poster
make test	unit tests (pytest)
make verify	checks of every committed result against its invariants

Table 4: Placement test. Each row compares the real wiring cost with 1000 permutations of cell-type positions.

analysis	real / permuted cost	z	permutations at or below real
soma positions, unweighted (primary)	0.459	-278.8	0 / 1000
synapse-weighted cost	0.320	-37.3	0 / 1000
permutation within compartment	0.728	-156.9	0 / 1000
synapse-centroid positions	0.277	-337.4	0 / 1000
brain subgraph only	0.679	-169.8	0 / 1000
nerve-cord subgraph only	0.741	-64.7	0 / 1000

exponential fit gives a different length constant in each setting: 63 μm within the brain, 128 μm within the nerve cord, and 208 μm across the two compartments. The neck crossings themselves have a median length of 597.6 μm , about three times that last constant, so they are made well beyond the range over which crossing between the compartments stays probable.

The real placement is not a local optimum. Of 2,000,000 proposed swaps within a compartment, 50,050 lowered the cost. Together they reduced it from 93,095 mm to 61,778 mm, a 33.64% reduction.

The search had not converged (the last 100,000 proposals still saved 0.33%), so this is a lower bound, and it ignores every physical constraint on where cell bodies and neuropils can lie.

The cable-length check supports cell-type positions as a between-class measure only (Figure 2). Across the 500 sampled neurons, skeleton cable length correlates with soma-to-output distance ($\rho = 0.428$, $p = 5.5 \times 10^{-24}$): descending and ascending neurons have more cable (medians 4,592 and 3,898 μm) than intrinsic neurons (1,428 and 1,713 μm). Within each superclass, however, $|\rho|$ is below 0.12.

Inside the neuropils, placement is economical too

This analysis was added after the registered tests, carries no registered hypothesis and is exploratory throughout.

Each cell type was assigned to the nearest of the 105 published neuropil surfaces. Cell types fall in 89 of them, at a median distance of 9.02 μm from the surface they are assigned to, and each connection lends half its length to the neuropil at each of its ends. Only 6.6% of the wire, 6,154 mm over 86,881 connections, stays inside one neuropil; the

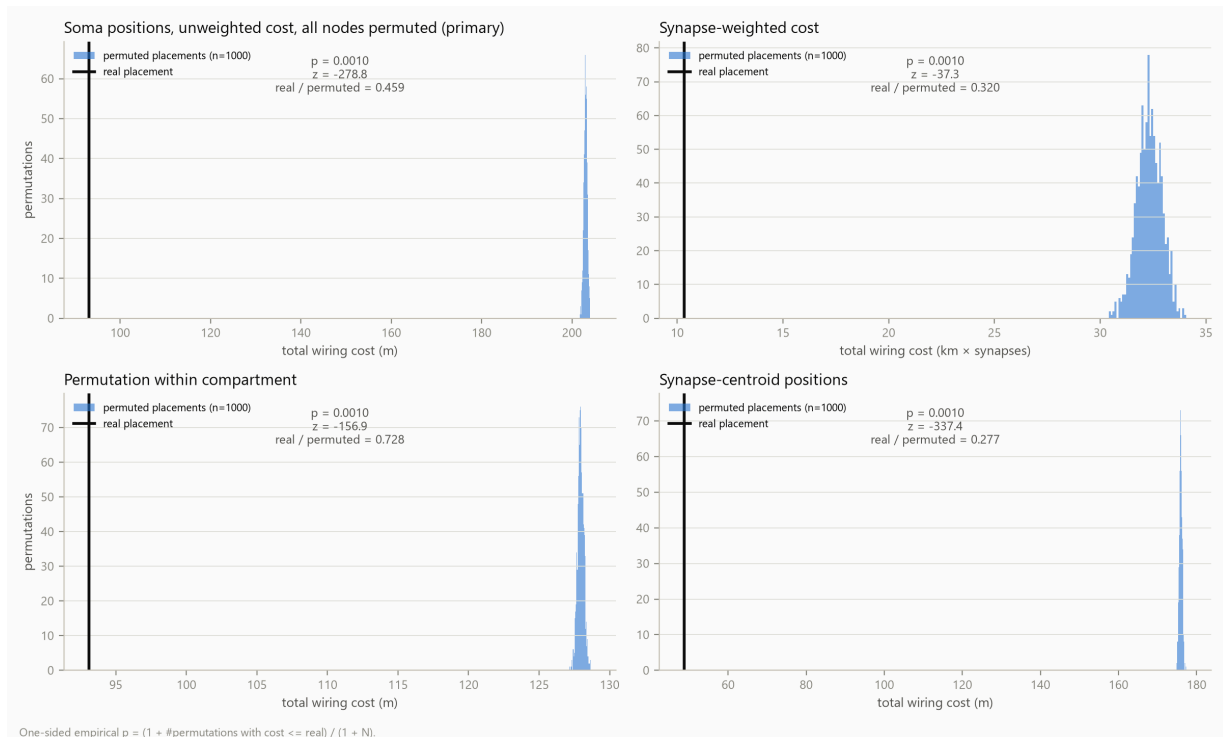


Figure 1: Real wiring cost against 1000 permutations of cell-type positions, for the primary and three secondary analyses.

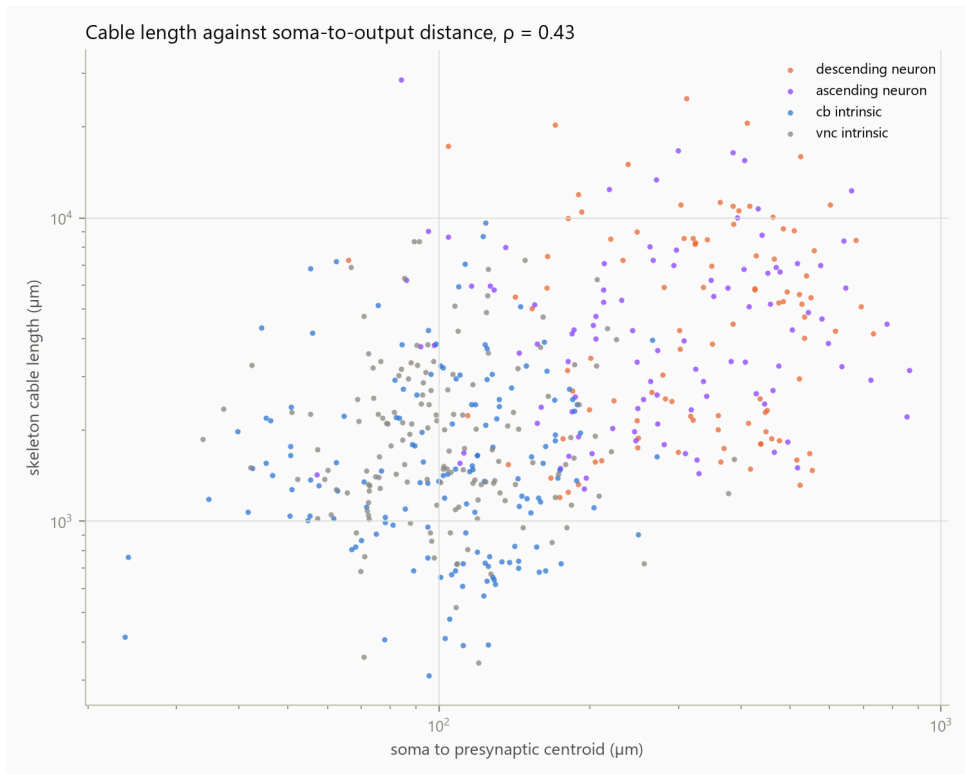


Figure 2: Skeleton cable length against soma-to-output distance for 500 sampled neurons, by superclass.

rest runs between them. The heaviest route between two regions joins the gnathal ganglia (GNG) to the abdominal neuromere (ANm), 1,586 mm over 2,101 connections.

The budget is not spread evenly over the regions (Table 5, Figure 3). GNG holds 10.1% of all wire, the largest share of any neuropil, ahead of ANm at 5.8%; the six leg neuropils of the nerve cord hold between 3.1% and 4.0% each.

Of the 89 neuropils, 48 hold enough cell types and internal connections to take that permutation test. All 48 come out below their permuted mean, and in 41 of them no permutation of the region's own cell types over its own positions was as short. The most economical are the mushroom body lobe gL(L) at 0.428 of its permuted mean ($z = -10.40$) and the two anterior optic tubercles, AOTU(R) at 0.516 and AOTU(L) at 0.536. The economy of the layout is therefore not only the separation of brain from nerve cord, nor of one neuropil from the next: it holds again inside nearly every region, over the positions that region already has.

Seven regions are the exception, and the two nearest the $\alpha = 0.05$ line fall on either side of it. The antennal lobes come closest to chance: AL(L) is at 0.974 of its permuted mean with 171 of 1000 permutations as short or shorter ($z = -1.00$, $p = 0.1718$), and AL(R) at 0.960 with 53 ($z = -1.62$, $p = 0.0539$), which misses α by four permutations in a thousand. The nerve-cord neuropil HTct(UTct-T3)(R) is at 0.886 with 49 ($z = -1.64$, $p = 0.0499$), and clears α by one. Nothing but those four permutations separates the two, and the pair should be read as one borderline result rather than as a difference between the regions. Of the remaining four, the accessory medulla AME(R) is at 0.939 with 231 ($z = -0.75$, $p = 0.2318$), while HTct(UTct-T3)(L) at 16 permutations ($p = 0.0170$), LOP(L) at 2 ($p = 0.0030$) and LOP(R) at 1 ($p = 0.0020$) are below α . None of the seven reaches the Bonferroni threshold for 48 regions, $\alpha = 0.05/48 = 0.00104$, which only the 41 regions at the smallest attainable p , $1/1001 = 0.00100$, clear. AL(L) holds 551 cell types and 3,139 internal connections, more than all but a few of the regions tested, and the two hemispheres agree. Inside the antennal lobes, the arrangement of cell types over their own positions saves little or no wire.

The wiring budget is concentrated, and its upper tail is not a power law

This analysis was also added after the registered tests and is exploratory.

The 490,884 connections hold 93,095 mm of wire between them, and they hold it unequally (Table 6, Figure 4). Sorted longest first, the longest 1% hold 4.6% of the budget, the longest 10% hold 30.7% and the longest quarter 52.2%; the Gini coefficient of the length distribution is 0.404. The longest tenth therefore carries 3.1 times its proportional share of the budget, and the longest quarter 2.1 times. That is uneven, but it is not the kind of concentration the rich-club literature describes: Towlson et al. (2013) report an eleven-neuron club holding 48% of the wiring cost of *C. elegans*, a

Table 5: Wire by neuropil. The twelve neuropils holding most wire, of the 89 that hold cell types. internal / permuted is the summed length of a neuropil's internal connections against the mean of 1000 permutations of which of its own cell types sits at which of its own positions.

neuropil	compartment	cell types	wire (mm)	share of wire	internal connections	internal / permuted	z
GNG	brain	1,918	9,445	10.1%	15,310	0.809	-33.39
ANm	nerve cord	1,108	5,420	5.8%	11,979	0.729	-23.94
AVLP(R)	brain	1,135	3,772	4.1%	5,881	0.863	-15.62
LegNp(T2)(R)	nerve cord	803	3,682	4.0%	3,764	0.885	-9.19
LegNp(T3)(R)	nerve cord	759	3,673	3.9%	2,603	0.894	-8.75
LegNp(T2)(L)	nerve cord	799	3,654	3.9%	3,791	0.895	-10.44
AVLP(L)	brain	1,075	3,572	3.8%	5,096	0.879	-11.18
LegNp(T1)(L)	nerve cord	667	3,192	3.4%	1,751	0.891	-7.72
LegNp(T3)(L)	nerve cord	636	3,082	3.3%	1,740	0.922	-5.72
SAD	brain	677	3,030	3.3%	1,842	0.682	-18.80
LegNp(T1)(R)	nerve cord	590	2,845	3.1%	1,493	0.885	-7.40
PB	brain	820	2,719	2.9%	2,277	0.767	-13.35

concentration onto a handful of cells rather than onto a tenth of the connections. The two figures are not measured over the same objects, and nothing in the length distribution here singles out a small set of connections in that way.

Neck-crossing connections are the longest and also the most uniform. Their median length, 597.6 μm , is four times the 146.3 μm within the brain, while their Gini, 0.151, is less than half the 0.343 within it. Crossing the neck costs about the same whichever pair of cell types does it.

The upper tail is heavier than an exponential, but a power law is not the best description of it. Fitted above 161 μm , over 231,054 connections, the exponent is 2.99. By normalized log-likelihood ratio the power law beats an exponential (46.81) and loses to both a lognormal (-45.52) and a truncated power law (-62.52), each comparison with $p < 0.0001$. The longest connection in the graph is 1,003.9 μm , against the 995 μm the somata span along their longest axis: the distribution is cut off by the extent of the nervous system itself, which is what the truncated fit picks up and the plain power law cannot.

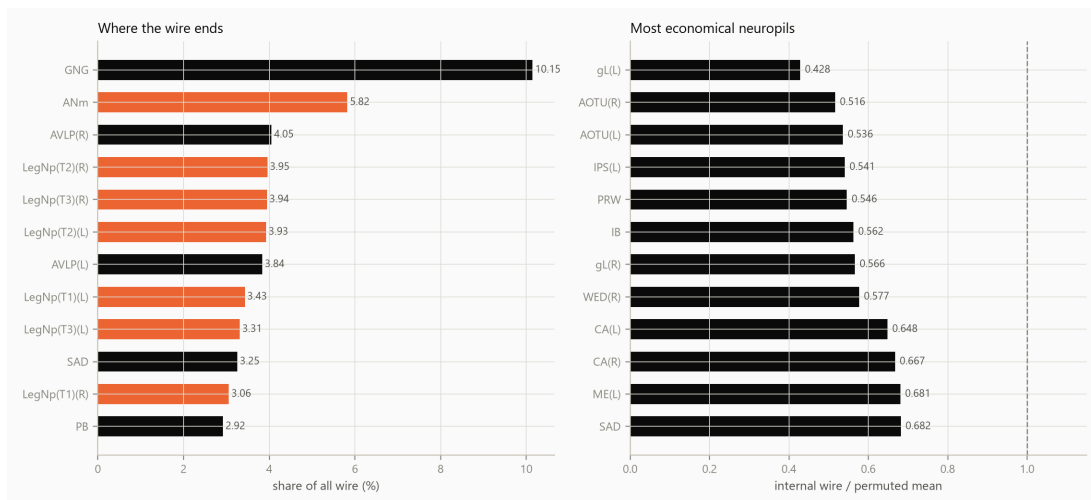


Figure 3: The wire held by the twelve neuropils that hold most of it, and how economically each is wired internally.

Table 6: Length and concentration by where a connection runs. The groups are views of the same 490,884 connections, not a partition: a connection between a brain type and a nerve-cord type that is neither descending nor ascending falls in none of the last three.

group	connections	wire (mm)	mean (μm)	median (μm)	99th percentile (μm)	Gini	longest 10%
all	490,884	93,095	189.6	154.3	808.8	0.404	30.7%
within the brain	319,431	49,540	155.1	146.3	544.7	0.343	23.6%
within the nerve cord	133,277	21,855	164.0	145.2	570.5	0.334	24.0%
across the neck	36,943	22,371	605.6	597.6	941.4	0.151	14.6%

Table 7: Wire by cell class, for the eight classes that own most of it. A connection is counted for the class at each of its two ends, so the shares add to more than one.

class	cell types	connections	wire (mm)	share of wire	mean length (μm)	median (μm)
central brain intrinsic	13,027	300,466	51,887	55.7%	172.7	152.0
nerve cord intrinsic	5,393	141,088	29,829	32.0%	211.4	158.1
ascending	1,089	57,516	20,906	22.5%	363.5	306.7
descending	951	49,647	19,013	20.4%	383.0	332.9
visual projection	681	31,050	4,866	5.2%	156.7	149.8
nerve cord sensory	328	18,765	2,730	2.9%	145.5	123.9
optic lobe intrinsic	529	17,415	2,338	2.5%	134.2	127.0
central brain sensory	342	13,299	1,574	1.7%	118.4	97.1

Central brain intrinsic types, 13,027 of the 23,073 cell types, touch connections holding 55.7% of the wire, and nerve cord intrinsic types 32.0% (Table 7). The 1,089 ascending and 951 descending types touch 22.5% and 20.4% of it, and their connections are the longest of any class with more than a thousand connections: mean 363.5 and 383.0 μm against 172.7 μm for central brain intrinsic types.

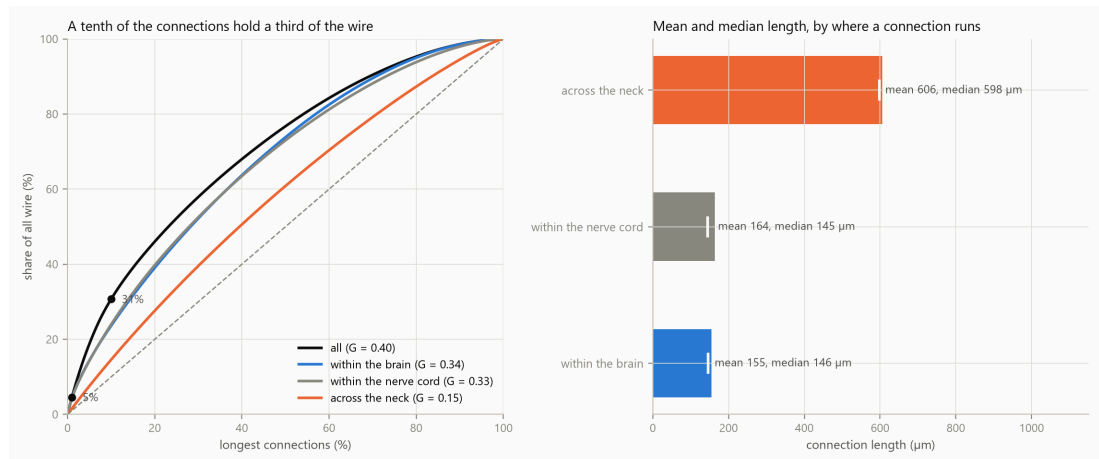


Figure 4: The Lorenz curve of the wiring budget, and the length of the connections behind it.

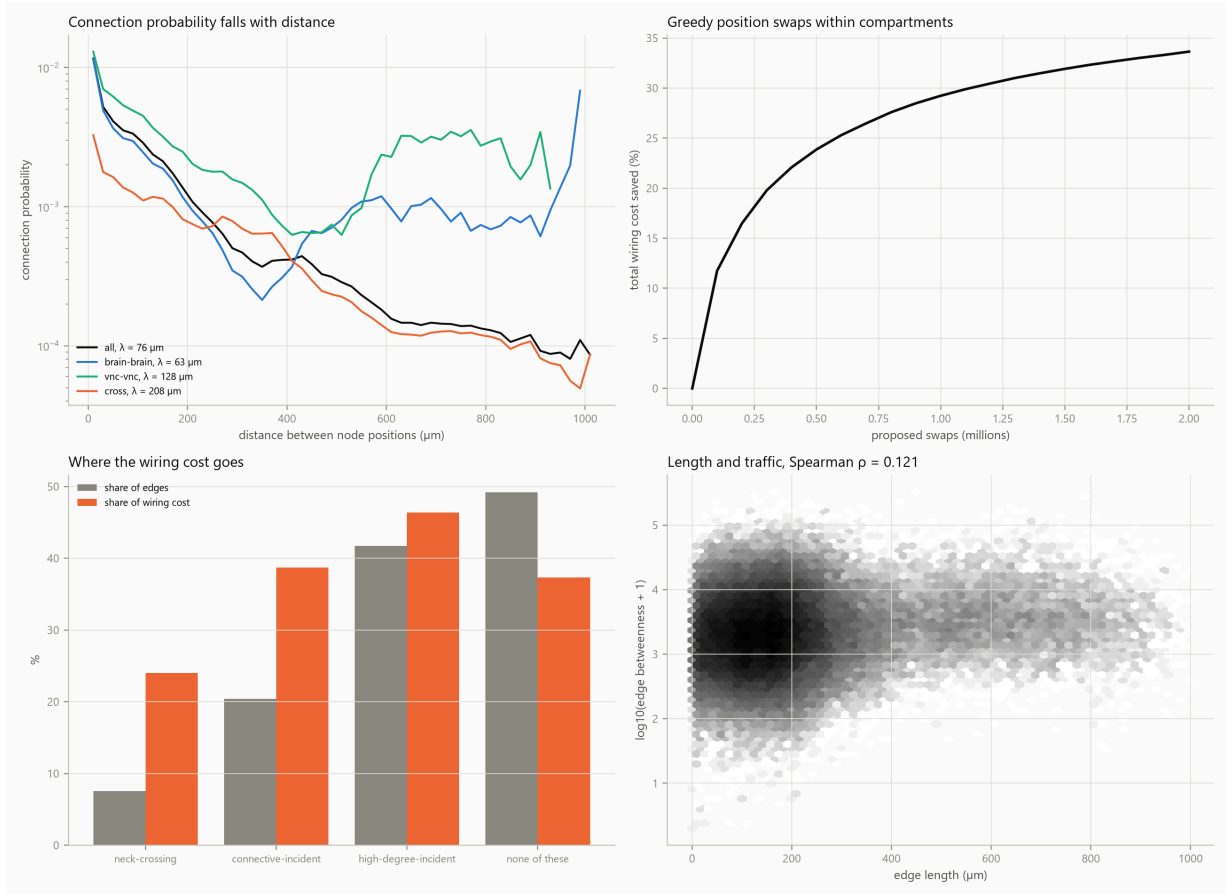


Figure 5: Wiring economy of the cell-type graph: connection probability against distance, cost saved by greedy swaps, the share of connections and of wiring cost by group, and connection length against edge betweenness.

The connective is expensive, and its partners are well connected

Neck-crossing connections are 7.5% of all connections and 24.0% of total wiring cost, with a mean length of 606 μm against 189.6 μm for all connections. Counting every connection incident on a connective cell type, the connective carries 20.4% of connections and 38.7% of cost.

Connections incident on high-degree cell types (total degree at least 67, the 90th percentile) are longer than other connections: mean 211 against 175 μm , median 161 against 150 μm (one-sided Mann-Whitney $p < 10^{-300}$). The gap in the middle of the distribution is therefore 11 μm , and the 41.8% of connections incident on a high-degree cell type carry 46.4% of the cost, 1.11 times their proportional share. The high-cost backbone reported in human and *C. elegans* connectomes is present here only in this weak form: high-degree cell types do attract longer wiring, but they do not concentrate the budget on it. Long connections also carry somewhat more traffic (Figure 5): connection length correlates with directed edge betweenness at $\rho = 0.121$, and at $\rho = 0.084$ among connections within one compartment.

Descending and ascending types are over-represented among the most connected cell types of the whole graph. At or above the 90th percentile of total degree lie 24.4% of descending types, 23.2% of ascending types and 8.8% of other types (odds ratio 3.22, Fisher $p = 1.5 \times 10^{-79}$). This is the expectation of Lin et al. (2024).

The connective's partners are also enriched for well-connected types. 14.4% of connective connections have a rich partner, against $10.2\% \pm 0.1\%$ when each connective cell type's partners are redrawn uniformly (ratio 1.40, $z = 33.9$, $p = 0.0010$).

The normalized rich-club coefficient of the whole graph exceeds all 100 randomized graphs from total degree 12 upwards, although at that degree the excess is negligible. The coefficient reaches 1.1 at degree 59, 1.5 at 88 and 2 at 172.

Rich-to-rich routing through the connective is modest and fragile

At the 1% input threshold, 7,454 two-step routes pass through a connective cell type from a rich partner on one side to a rich partner on the other. That is 1.103 times the mean of 1000 layer-preserving rewirings ($6,760 \pm 143$, $z = 4.8$, $p = 0.0010$), so the primary hypothesis is supported.

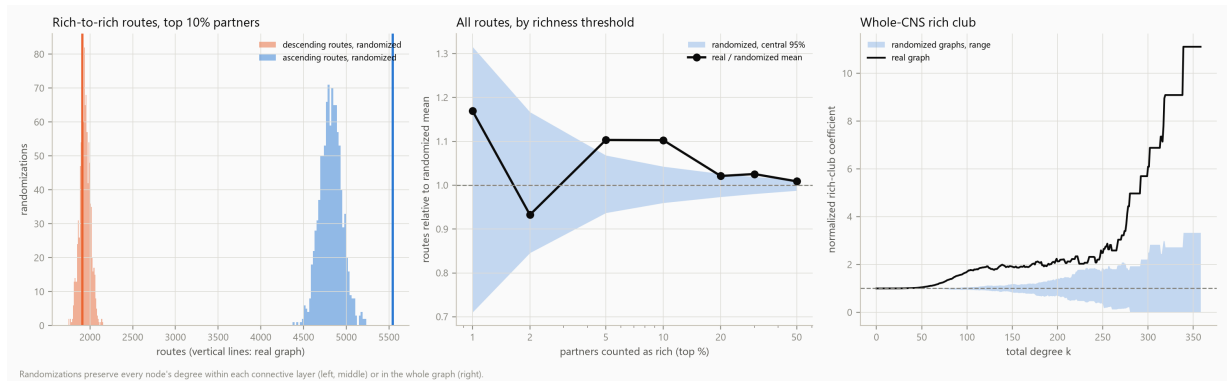


Figure 6: Rich-club structure of the connective. Rich-to-rich routes against layer-preserving rewirings, by richness threshold, and the normalized rich-club coefficient of the whole graph.

At this threshold the excess is carried entirely by ascending routes (ratio 1.150, $p = 0.0010$); descending routes are at chance (0.984, $p = 0.71$). It also depends on how “rich” is defined (Figure 6). R was computed at seven richness thresholds. It is significant when the top 5%, 10% or 30% of partners count as rich ($p = 0.0020$, 0.0010 and 0.0040) and not when the top 1%, 2%, 20% or 50% do. Only the top 10% carries the registered hypothesis and the other six are exploratory, but all three significant values also fall below a Bonferroni threshold across the seven, $\alpha = 0.0071$, so the pattern is not an artefact of trying several definitions.

The result depends more sharply on the input threshold, and the pre-registered robustness hypothesis failed because of it (Table 8):

The route enrichment grows as weak connections are dropped and disappears when they are added. Rich-to-rich routing through the connective is therefore a property of its strongest connections, not of its wiring as a whole.

The ascending–descending asymmetry does not survive the sweep either. At 0.5% the significant direction is the descending one (1.042, $p = 0.0090$) and the ascending is not significant (0.979, $p = 0.89$), the reverse of the 1% result. At 2% the ascending excess is larger (1.533) while the descending ratio falls further below one (0.857, $p = 0.99$). At 5% both directions are significant (1.311, $p = 0.042$ and 1.725, $p = 0.0010$). The direction of the asymmetry therefore changes with the threshold, and only the 1% figures were registered. That the excess is ascending is a statement about the 1% graph and cannot be read as a property of the connective.

Placement economy and the over-representation of connective types among hubs hold at every threshold, the hub odds ratio falling steadily from 3.99 to 1.33 as weak connections are dropped.

The connective buys less flow per unit of wire than ordinary wiring, except from brain to nerve cord

The intact graph supports 8,731 edge-disjoint sensory-to-motor paths. Cutting the 36,943 neck-crossing connections (22,371 mm of wire) removes 963 of them. Random non-connective sets of the same total length, about 153,000 shorter connections each, remove $2,689 \pm 41$ (ratio 0.36, $z = -42.5$, $p = 1.0$).

Table 8: Robustness to the input threshold. Placement and hub over-representation hold at every threshold; rich-to-rich routing does not hold at 0.5%, and the split between descending and ascending routes holds nowhere.

input threshold	connections	real / rewired routes	p	descending ratio (p)	ascending ratio (p)	hub odds ratio	placement ratio
0.5%	942,400	1.002	0.42	1.042 (0.0090)	0.979 (0.89)	3.99	0.460
1%	490,884	1.103	0.0010	0.984 (0.71)	1.150 (0.0010)	3.22	0.459
2%	216,862	1.391	0.0010	0.857 (0.99)	1.533 (0.0010)	2.22	0.455
5%	52,217	1.605	0.0010	1.311 (0.042)	1.725 (0.0010)	1.33	0.439

Table 9: Value of the neck-crossing connections. Null sets are 1000 random sets of non-connective connections matched on total length (primary) or on count; the last row is a single deterministic set.

comparison	flow lost, connective	flow lost, null mean	ratio	p
all flow, equal length (primary)	963	2,689	0.36	1.0
brain sensory → nerve-cord motor, equal length	3,759 of 6,423	2,052	1.83	0.0010
nerve-cord sensory → brain motor, equal length	242 of 2,228	555	0.44	1.0
all flow, equal connection count	963	648	1.49	0.0010
brain → nerve cord, equal connection count	3,759	464	8.10	0.0010
all flow, longest non-connective connections	963	962	1.00	—

The primary value hypothesis is not supported: per unit of wire, neck-crossing wiring carries less sensory-to-motor flow than ordinary wiring. No sensory-motor pair is disconnected by the cut. Every pair the connective joins is also joined by a path that avoids the neck-crossing connections.

The loss is concentrated in one direction (Table 9, Figure 7):

Cutting the neck removes 59% of the flow from brain sensory to nerve-cord motor cell types, 1.83 times what random wiring of equal length removes. In the other direction it removes 11%, 0.44 times the random loss.

Per connection, neck-crossing connections carry more flow: they remove 1.49 times the flow of an equal number of random non-connective connections.

The comparison that most directly answers whether the connective is special is the last row. The 84,008 longest non-connective connections, each at least 207 μm long, together match the connective's total length. Removing them costs 962 units of flow, one fewer than the connective. On this measure the connective is worth what long wiring of its length is worth. The exception is its brain-to-nerve-cord direction, where the longest connections remove only 1,122 units against its 3,759.

Silencing every connection of the descending and ascending cell types, not only the neck-crossing ones, removes 2,118 units of flow, 0.49 times a length-matched null. It also removes 6,202 of the 6,423 units from brain to nerve cord and 1,543 of the 2,228 from nerve cord to brain, 1.85 and 1.72 times the null. What crosses between brain and nerve cord depends almost entirely on these neurons, the bottleneck Berg et al. (2026) described.

Across cell types, the price of the connective does not predict its value

Two hypotheses were registered at the level of the single cell type. H10 asked whether a connective cell type's wire length correlates with the flow it carries, separately among descending and among ascending types. H10b asked whether any such correlation survives control for the number of neck-crossing connections the type has. Neither is supported, and they fail differently on the two sides of the neck.

For a single descending or ascending cell type, removing its neck-crossing connections usually costs no flow at all: this holds for 622 of 911 descending and 899 of 1,010 ascending types. The remaining wiring supports the same maximum flow, so most individual members of the connective are, on this measure, individually dispensable.

Among descending types, price and value are correlated ($\rho = 0.231$, $p = 7.6 \times 10^{-13}$). Controlling for the number of neck-crossing connections leaves almost nothing (partial $\rho = 0.044$, $p = 0.09$): at a given number of connections, longer wiring buys no more flow. Among ascending types price and value are unrelated ($\rho = -0.005$, $p = 0.56$) and the partial correlation is negative (-0.122).

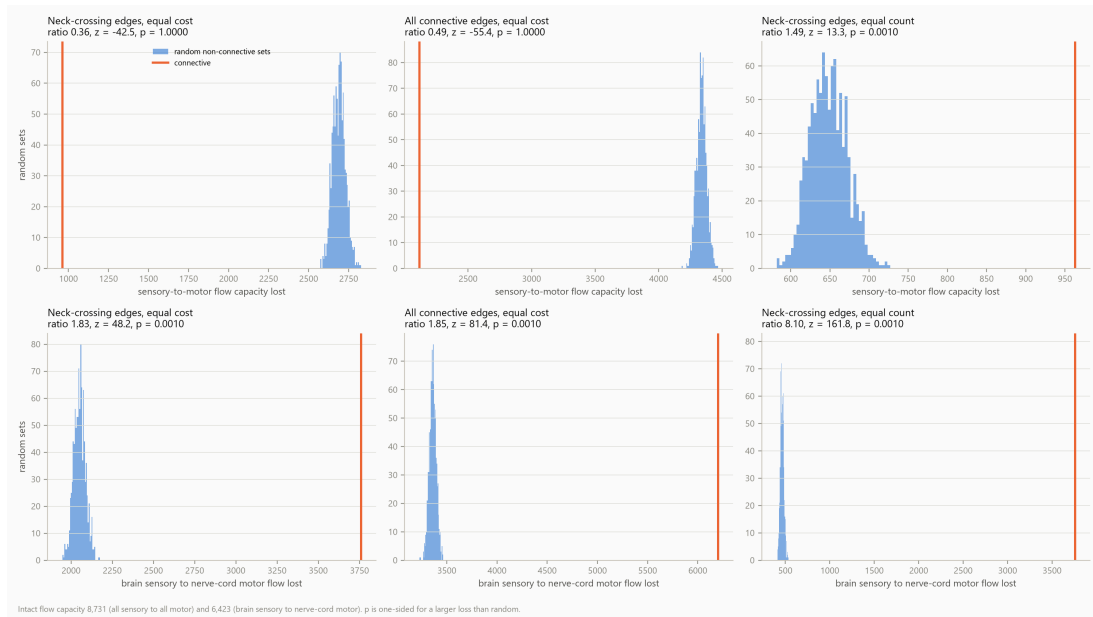


Figure 7: Sensory-to-motor flow lost when the neck-crossing connections are cut, against 1000 cost-matched and count-matched sets of non-connective connections.

What separates the two directions is how strongly the number of connections predicts value at all. Among descending types the rank correlation between connection count and value is 0.227; among ascending types it is 0.027. The descending price–value correlation, 0.231, is almost exactly the correlation that connection count already supplies, which is what partialling removes. On the ascending side there is nothing for price to inherit: neither wire nor connection count predicts the flow an ascending type carries in its own direction.

The ranked extremes make the failure concrete (Table 10, Figure 8). Of the fifteen most expensive ascending types, thirteen carry no nerve-cord-to-brain sensory-to-motor flow of their own at all. The most expensive after the two copies of AN07B004 is AN09B004, with 103.9 mm of neck-crossing wire over 123 connections on the left and 91.0 mm over 110 on the right; removing those connections costs nothing in either case. AN06B009, ANXXX027, AN00A006, AN08B018, AN05B101 and AN02A002 are the same, each spending between 74 and 99 mm for no flow. The two exceptions are the copies of AN07B004 itself, the most expensive cell type anywhere in the connective, with 316 and 313 neck-crossing connections and 146.2 and 137.4 mm of wire; each carries 2 units of flow, 0.01 per millimetre.

Table 10: The most expensive cell types of the connective in each direction, by the summed length of their neck-crossing connections, with the flow capacity lost when those connections alone are cut. Every cell type’s figures are in results/connective_price.csv.

direction	cell type	side	connections	price (mm)	flow value
descending	DNg98	R	177	122.8	16
descending	DNg98	L	172	114.6	16
descending	DNae009	L	93	68.8	0
descending	DNae009	R	93	67.7	0
descending	DNg08	L	88	55.5	0
descending	DNg08	R	82	53.3	0
ascending	AN07B004	R	316	146.2	2
ascending	AN07B004	L	313	137.4	2
ascending	AN09B004	L	123	103.9	0
ascending	AN06B009	R	184	98.9	0
ascending	ANXXX027	L	112	95.9	0
ascending	AN09B004	R	110	91.0	0

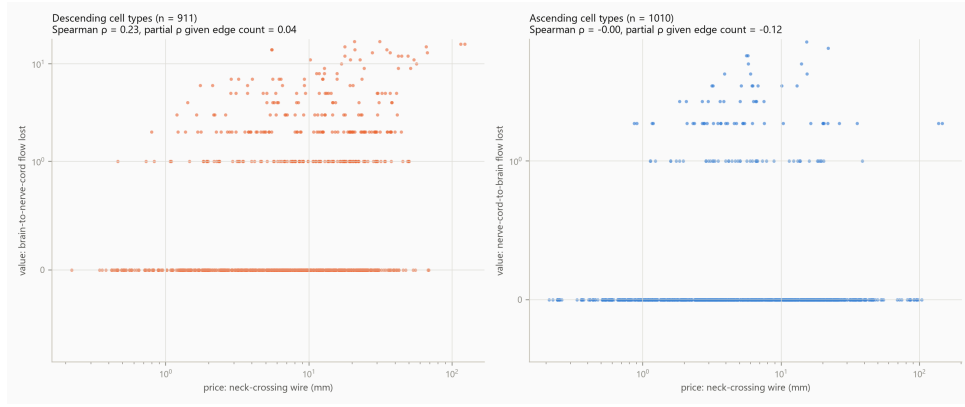


Figure 8: Price and flow value of individual descending and ascending cell types.

The descending side is less extreme but shows the same thing at the top. DN_g98 is both the most expensive descending type, at 122.8 and 114.6 mm on the two sides, and among the most valuable, at 16 units each. Immediately behind it, DN_{ae}009 spends 68.8 and 67.7 mm on 93 neck-crossing connections per side and carries no flow, and DN_g08 spends 55.5 and 53.3 mm on 88 and 82 connections and also carries none. Four of the fifteen most expensive descending types carry zero and three more carry one.

Value per millimetre runs the other way entirely. The best-value descending type, DN_{ge}002, carries 14 units on 5.5 mm over 14 connections, 2.54 units per millimetre, nearly twenty times DN_g98's 0.13; DN_{pe}034 and DN_{pe}036 carry between 14 and 17 units each on 17.8 to 34.9 mm. On the ascending side AN_{xxx}191 carries 7 units on 5.6 mm and AN₀₇B042 7 units on 5.7 mm, against AN₀₇B004's 2 on 146.2 mm. The cell types that carry the most flow across the neck are not the ones that spend the most wire on it.

That is the substance of both failures. Price tracks value among descending types only through the number of connections, and not at all among ascending ones, where the largest expenditures in the whole connective buy nothing this measure can see. What a zero means here is limited: flow capacity counts edge-disjoint paths, so it records only that the rest of the wiring still supports the same maximum flow once that type's neck crossings are gone, not that the type is idle or that the animal could spare it.

A distance and cell-class model reproduces the rich-to-rich routes but little else

Distance alone explains connections moderately well, with McFadden pseudo- R^2 0.165 and cross-validated AUC 0.786. The other terms add fit as follows (Table 11):

Model G carries two distance terms. The log-odds of a connection fall by 0.248 per 100 μ m of distance and by a further 0.705 per unit of log distance, so the penalty is steepest over the first few tens of micrometres and flattens beyond them; a shared compartment adds 0.698. The four largest class-pairing terms all involve descending neurons.

Model G reproduces 2 of the 13 compared properties: the maximum in-degree and the number of rich-to-rich routes through the connective (Figure 9). Its synthetic graphs have 6,636 such routes (central 95% 5,540–7,600) against 7,454 real. The model knows nothing of the real degree sequence. The route count that exceeds degree-preserving rewiring by 10% therefore lies within the range that distance, compartment and cell class produce on their own.

Model G+deg matches the out-degree distribution far better (Kolmogorov-Smirnov distance 0.039 against 0.351). It reproduces only reachable pairs, however, and makes rich-to-rich routes 2.1 times too frequent.

Neither model reproduces the local structure of the real graph, which has 21 and 16 times the reciprocity of their graphs and 16 and 8 times their transitivity, as expected when every ordered pair is drawn independently. Neither reproduces

Table 11: Fit of the nested generative models.

	model	pseudo- R^2	cross-validated AUC
	distance only	0.165	0.786
1.	compartment	0.165	0.787
1.	superclass pairing (model G)	0.276	0.847
1.	source and target degree (model G+deg)	0.381	0.899

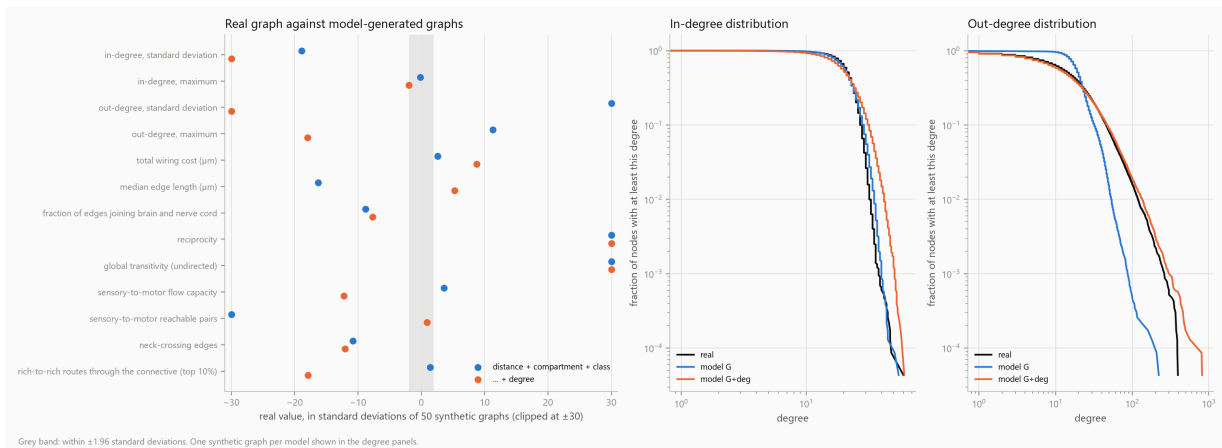


Figure 9: Thirteen properties of the real graph against 50 synthetic graphs from each generative model.

total wiring cost, median connection length, the number of neck-crossing connections (39,241 under model G against 36,943) or flow capacity (8,365 against 8,731).

Degree says nothing about where a cell type sits, but a type sits among its partners

This analysis and the four that follow it were run after all thirteen registered tests had been computed. None carries a registered hypothesis, none had a direction fixed in advance, and every p-value below is uncorrected.

A heavily connected cell type pays for its position on every one of its connections, so wiring economy predicts that the best connected types sit centrally. They do not. Over all 23,073 cell types the rank correlation between total degree and distance from the centroid of the types in the same scope is -0.043 . Split by compartment the two halves do not agree in sign: $+0.018$ over the 16,052 brain types and -0.144 over the 7,021 nerve-cord types, the largest of the six correlations taken and still small. Distance is measured from the centroid of the scope rather than of the whole specimen, because a shared centre would rank a type by which compartment it belongs to rather than by where it sits inside it; the median type lies $297.1\ \mu\text{m}$ from the centre of the whole graph, $162.5\ \mu\text{m}$ from the centre of the brain and $157.8\ \mu\text{m}$ from the centre of the nerve cord. The summed length of a type's connections behaves no differently: $\rho = 0.002$ over all types, -0.098 in the brain and -0.054 in the nerve cord. How well connected a cell type is says next to nothing about how far from the middle of its compartment it sits.

The second test is not a null (Figure 10). Each type keeps its degree and draws that many partners uniformly at random from the other 23,072, with replacement and never itself. Over 1000 such draws the mean distance from a type to the centroid of its real partners is $127.3\ \mu\text{m}$ against $320.7\ \mu\text{m}$ for random partners, a ratio of 0.397 ($z = -622.7$, no draw as near, $p = 0.0010$). Within the brain it is 120.6 against $259.3\ \mu\text{m}$ and within the nerve cord 142.6 against $460.9\ \mu\text{m}$, ratios of 0.465 and 0.309 . Of the 23,070 cell types with at least one connection, 95.4% sit nearer their own partners than their own null mean. The type furthest below its own null is the nerve-cord type SNppxx on the right, $18.4\ \mu\text{m}$ from the centroid of the 402 partners it contacts, 29.1 null standard deviations below its own mean.

The two results are compatible, and together they say what kind of economy this is. Placement economy at cell-type resolution is local rather than radial: a type sits among the cells it contacts, and being well connected does not pull it towards the middle of its compartment.

The two sides of the body hold the same wire

Exploratory, with no registered hypothesis and uncorrected p-values, like the four analyses either side of it.

Of the 89 neuropils that hold cell types, 72 form 36 left-right pairs and carry 75.4% of the wire between them. Within a pair the two sides hold wire within a factor of 1.13 of each other at the median and 1.30 on average, and over all 36 pairs the right side holds a geometric mean of 1.132 times the wire of the left (Figure 11). The widest gap is LA, where the right copy holds $6.80\ \text{mm}$ over four cell types against $0.50\ \text{mm}$ over one, a factor of 13.5; the next widest are FLA at 2.3 and AME at 2.1, both again in favour of the right.

Two nulls, because the question has two halves. If the two hemispheres are wired alike then the sign of a pair's log ratio is arbitrary, so negating each of the 36 ratios at random gives the null distribution of their mean. The observed mean is -0.124 against a null spread of 0.0836 ($z = -1.49$, two-sided $p = 0.109$ over 1000 sign flips). There is no side bias in the wiring budget that this test can separate from chance. The second null matches each left neuropil to a right neuropil drawn at random instead of to its own counterpart: the observed mean absolute log ratio is 0.260 against 1.869 ± 0.179 , and none of 1000 matchings is as small ($p = 0.0010$). Counterparts are far more alike than arbitrary neuropils, which is

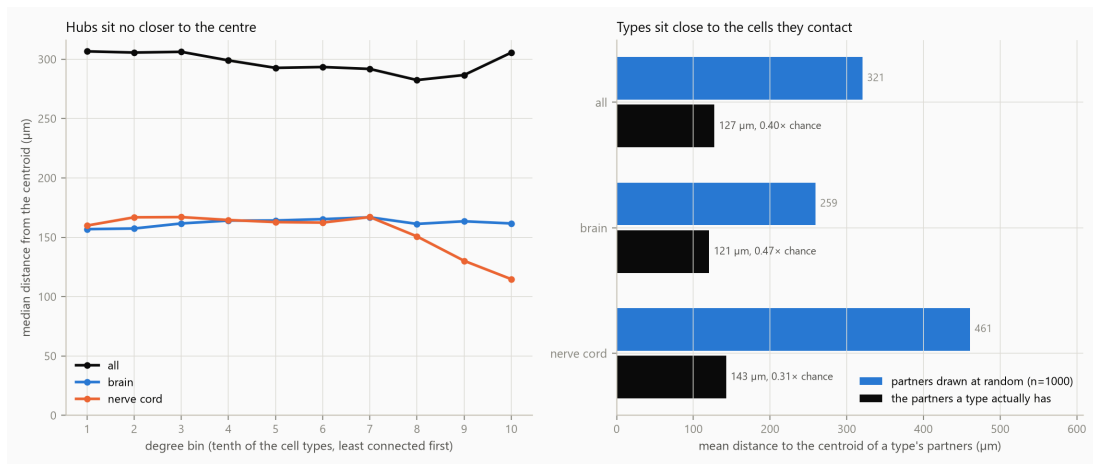


Figure 10: Median distance from the centre of the compartment against degree, and each cell type's distance to the centroid of its own partners against degree-matched random partners.

what the naming asserts and is worth confirming, but the null is weak in a specific way. It sets no scale for how alike two copies of one structure ought to be, so it can rule out only the case where the pairing carries no information at all, and it cannot say whether a factor of 1.13 between two copies is small.

Placement economy is as symmetric as the budget. Twenty pairs have an internal cost ratio on both sides, from the within-neuropil permutation test above. Their mean left-minus-right difference is -0.0074 against a sign-flip null spread of 0.0132 ($z = -0.56$, $p = 0.607$), and the mean absolute difference is 0.0389 on ratios that all sit below one. Whatever placement economy a neuropil has, its opposite copy has about the same amount of it.

Seventeen neuropils take no part in these tests and are named here rather than dropped. Twelve carry no hemisphere suffix and hold 24.6% of the wire between them: GNG, ANm, SAD, PB, IB, PRW, IntTct, LTct, CV-anterior, CRN, FB and EB, a set that includes the largest single holder of wire in the specimen. Five are one side of a structure whose other side holds no cell type in this reconstruction, and hold 29.2 mm, 0.03% of the wire, between them: bL(R), AB(L), SCL(R), VES(R) and b'L(L).

On neither test is the observed asymmetry larger than expected, and the direction of the differences is not consistent enough across pairs to register. The left-right difference in this specimen's wiring budget is a null result.

A long connection carries about as many synapses as a short one

Exploratory, with no registered hypothesis and uncorrected p-values. The result is a null, and it carries the section.

The 490,884 connections hold 78,192,944 synapses over 93,095 mm of wire, 0.840 synapses per micrometre. Sorted by length into ten equal groups, the shortest tenth buys 13.37 synapses per micrometre and the longest 0.11, a factor of 117 (Figure 12). Almost all of that fall is the length in the denominator.

Length and synapse count are all but uncorrelated. Spearman's ρ between a connection's length and the number of synapses it carries is -0.054 over all 490,884 connections, and Pearson's r between the logarithms of the two is -0.063 .

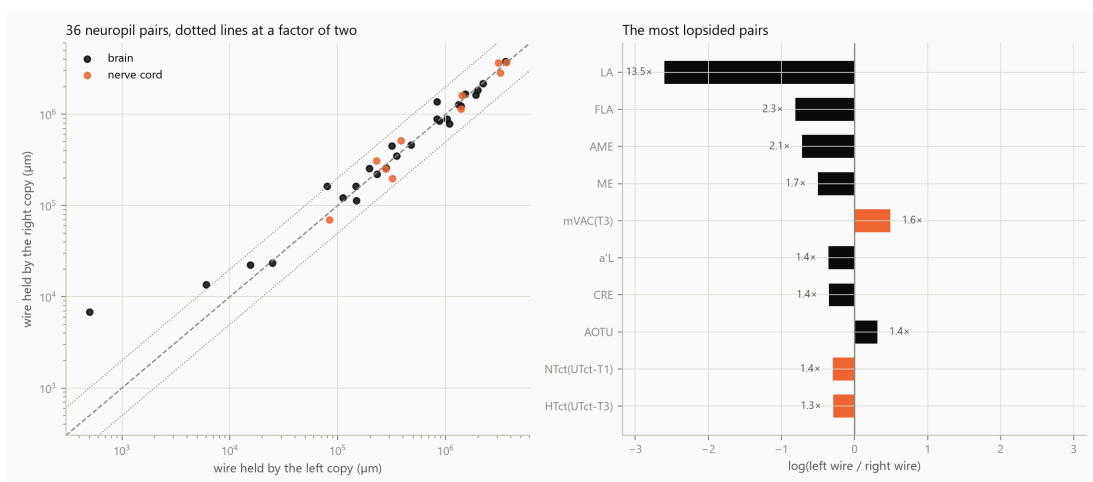


Figure 11: The wire held by the two copies of each of the 36 paired neuropils, and the log ratios of the most lopsided pairs.

With this many connections a correlation that small still returns a p-value at the floor of double precision, which is a statement about the number of connections and not about the size of the effect; it is reported here as one and nothing is inferred from it. The median connection carries 40 synapses in the shortest decile and 37 in the longest, a ratio of 0.93. The decile means fall much further, from 349.8 to 66.5, and the distance between mean and median says those means are set by a small number of very heavy connections rather than by the typical one.

The numerator is flat, but it is not perfectly flat, and the decile table says by how much. Dividing each decile's density by the density it would show if synapse count did not depend on length at all leaves a residual that runs from 2.20 in the shortest decile to 0.42 in the longest, non-monotonically: deciles two to five sit between 1.09 and 1.38 and deciles seven to ten between 0.42 and 0.56. A connection under 46 μm long therefore does carry more synapses than its length alone would predict, by a factor of about two, and one over 360 μm rather fewer. That is a 5.3-fold spread against the 117-fold fall in the raw density, and it is carried by the means rather than the medians: the shortest decile averages 349.8 synapses per connection against 159.3 over the whole graph, while its median is 40 against 40.

The same holds between the two groups of connections that differ most in length. Neck-crossing connections average 606 μm against 156 μm elsewhere. They hold 24.0% of the wire and 2.9% of the synapses, so they buy 0.103 synapses per micrometre against 1.073 elsewhere, a factor of 10.4, and yet they carry a median of 37 synapses against 40. Within the neck-crossing connections the correlation between length and synapse count is +0.089 and within the rest it is -0.048. Both are negligible, so inside each group as well as across the whole graph the price per micrometre is set by the length and not by what the connection carries.

Synapses per micrometre is therefore a ratio whose numerator is flat and whose denominator spans the length distribution, and its decline with length is closer to arithmetic than to a discovered relationship. The finding worth stating is the flat numerator. A cell type pays for reach by the micrometre and receives, for each connection it makes, a synaptic contact of much the same size wherever the partner sits. The null also bears on the value tests above, where flow capacity counts a connection once whatever its weight: treating connections as unweighted discards no systematic relationship between a connection's length and its synaptic load, because there is none to discard.

The regions form a dense network that is not small-world by wire

Exploratory, with no registered hypothesis and uncorrected p-values.

Collapsing the cell-type graph onto the neuropils gives a network of 89 regions joined by 2,698 weighted edges, 68.9% of the 3,916 pairs that could be joined. 93.4% of the wire runs between two regions, 86,941 mm of it, and the remaining 6,154 mm stays inside one. The gnathal ganglia are the largest hub, holding 16,058 mm of wire to 85 of the other 88 regions, 9.2% of all between-region wire, and the fifteen largest regions hold 56.8% of it between them. Degree separates the regions far less than wire does: the median region has 69 partners and the least connected has two.

The network is not small-world by wire (Table 12, Figure 13). Against 1000 degree-preserving rewirings with the wire redealt it is 1.099 times as clustered, but its weighted paths are 10.1 times longer rather than shorter, which puts the wire-weighted small-world ratio at 0.109, far below the value a small-world network would give. The second null, which keeps the real topology and permutes only the wire over it, reproduces the same direction at 9.9 times its mean, so the long weighted paths follow from heavy pairs of regions sitting together rather than bridging the network. The

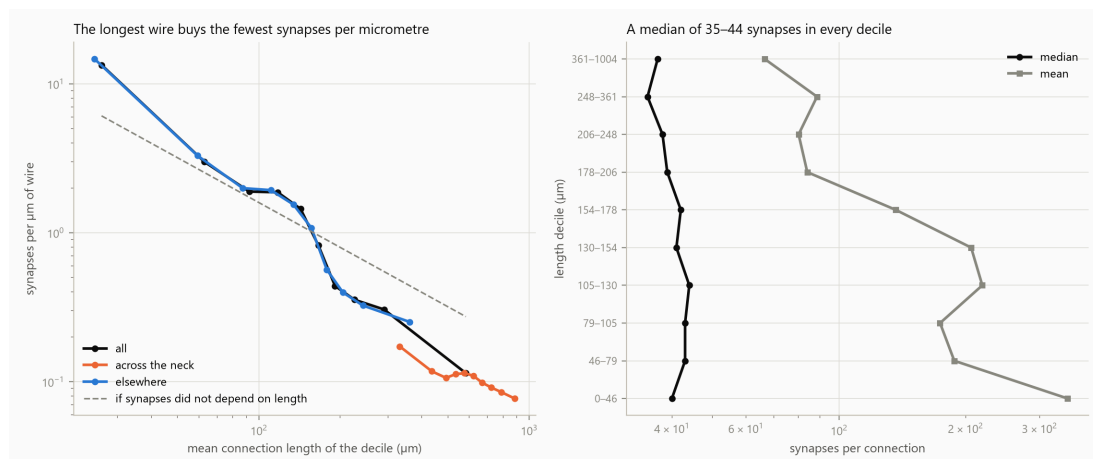


Figure 12: Synapses bought per micrometre of wire by length decile, and the synapses carried by a single connection in each decile.

Table 12: The 89-region network against two nulls of 1000 draws each, both preserving every region's number of partners. Every ratio has a two-sided p of 0.0020 except the two the weight-shuffled null reproduces by construction, which have p = 1.

metric	real	rewired null	real / rewired	weight-shuffled null	real / shuffled
clustering	0.872	0.862 ± 0.002	1.011	0.872 ± 0.000	1.000
path length	1.319	1.312 ± 0.001	1.005	1.319 ± 0.000	1.000
weighted clustering	0.947	0.862 ± 0.005	1.099	0.872 ± 0.005	1.086
weighted path length	292.4	29.0 ± 13.9	10.100	29.6 ± 14.4	9.888

unweighted version of the question is degenerate at this density and cannot answer it either way: with 68.9% of pairs already joined, unweighted clustering is 0.872 against a null mean of 0.862 and the mean path is 1.32 steps against 1.31.

Louvain on the wire-weighted region graph returns exactly two communities, modularity 0.229, and all 100 random seeds return the same partition. They follow the split between brain and nerve cord and nothing else. The adjusted Rand index against compartment is 0.626 over all 89 regions; against hemisphere it is -0.010 over the 77 regions carrying a side suffix and against thoracic segment 0.000 over the 18 regions naming one, both at chance. Of the 36 neuropils present on both sides only ICL has its two copies in different communities, and the leg neuropils of all three thoracic segments sit together. The wire that joins a region to its mirror image, or to its neighbour one segment along, does not separate either from the rest of its compartment.

The correspondence with the compartment is strong but not exact. The second community holds all 23 nerve-cord neuropils together with nine brain neuropils — GNG, SAD, IB, ICL(R), CV-anterior, CRN, FLA(R), FLA(L) and VES(R) — the gnathal and ventral-posterior regions that lie against the neck. On wire alone they group with the nerve cord rather than with the rest of the brain. These two communities are a partition of anatomy by wire; nothing here identifies a functional module.

Per micrometre, short connections buy more connectivity than long ones

Exploratory, with no registered hypothesis and no direction fixed in advance.

Connections were removed cumulatively from one end of the length distribution and the graph left behind measured at each of 18 points (Figure 14). The whole graph has a sampled global efficiency of 0.2152 and holds 23,070 of its 23,073 cell types in one weakly connected component.

At the reference point, a quarter of the wire removed, the two ends of the distribution are not comparable (Table 13). Taking that 23,274 mm from the long end costs 35,849 connections, 7.3% of them, and leaves efficiency at 93.5% of the whole graph's. Taking the same wire from the short end costs 255,849 connections, 52.1%, and leaves it at 71.7%. The first removal costs 6.5% of the starting efficiency and the second 28.3%, a factor of 4.4 for the same wire. Efficiency falls to half its starting value once 38.9% of the wire has been taken from the short end, and not within the 60% sampled from the long end, where it still stands at 69.5%.



Figure 13: The regions holding the most wire and how many partners each reaches, the wire between all 89 regions ordered by community, and the wire-weighted small-world measures against both nulls.

Table 13: What removing a quarter of the wire, 23,274 mm, costs from each end of the length distribution, against the same number of connections removed at random. The random row is the mean of five draws.

schedule	connections removed	share of connections	efficiency	share of the starting value	largest component
longest first	35,849	7.3%	0.2013	93.5%	99.99%
shortest first	255,849	52.1%	0.1544	71.7%	98.94%
at random, matched on count	35,849	7.3%	0.2105	97.8% $\pm$ 0.2%	99.99%

The reason is arithmetic rather than subtle. The median connection is 154.3 μm long against a mean of 189.6 μm , so the same budget taken from the short end removes 7.1 times as many connections. Wire spent on the long tail is not the cheapest way to keep the graph short-pathed.

Matched on the number of connections instead, the comparison reverses, and the countervailing result is as clear as the first. Removing the 35,849 longest connections leaves efficiency at 93.5%, while removing the same number at random leaves 97.8% $\pm$ 0.2% over five draws. Individually the longest connections are worth more to the graph than typical ones. They are simply not worth their length.

The third measure returns nothing, and its silence is worth as much space as the rest. None of the three schedules breaks the graph apart over the range sampled. The largest weakly connected component holds 99.99% of the cell types after a quarter of the wire is taken from the long end, 98.94% after the same wire is taken from the short end and 99.99% after the matched number of random removals. At the far end of the range, with 60% of the wire gone, the long-end schedule has cost 31% of the efficiency while leaving 99.90% of the types in one component, and the short-end schedule has cost 86% of the efficiency while still leaving 79.5% of them in one. Component size is a blunt instrument at this density: efficiency has collapsed while the graph is still almost entirely one piece. That is a null for the component measure rather than evidence that the graph is robust in any useful sense.

Efficiency here is a graph measure over shortest paths in a static wiring diagram. It says what removal does to the lengths of those paths and nothing about signalling.

Sensitivity and robustness checks

- **Input threshold.** Placement holds at every threshold tested, with a cost ratio between 0.460 and 0.439, and hub over-representation holds with an odds ratio falling from 3.99 to 1.33. Rich-to-rich routing fails at 0.5% and its ascending-descending split reverses between thresholds, so H11 is not supported (Table 8).
- **Cost measure.** The placement result holds under the synapse-weighted cost (0.320), under synapse centroids as positions for every cell type (0.277) and within the brain and nerve cord separately (0.679 and 0.741), as well as under the primary unweighted cost (Table 4).
- **Richness definition.** Three of seven richness thresholds give a significant route excess, and all three stay below a Bonferroni threshold across the seven ($\alpha = 0.0071$). The primary 10% threshold was registered in advance.
- **Null construction for value.** The three nulls disagree, and all three are reported: against equal total length the connective removes 0.36 times the flow, against an equal number of connections 1.49 times, and against the longest non-connective connections of the same total length 1.00 times (Table 9).

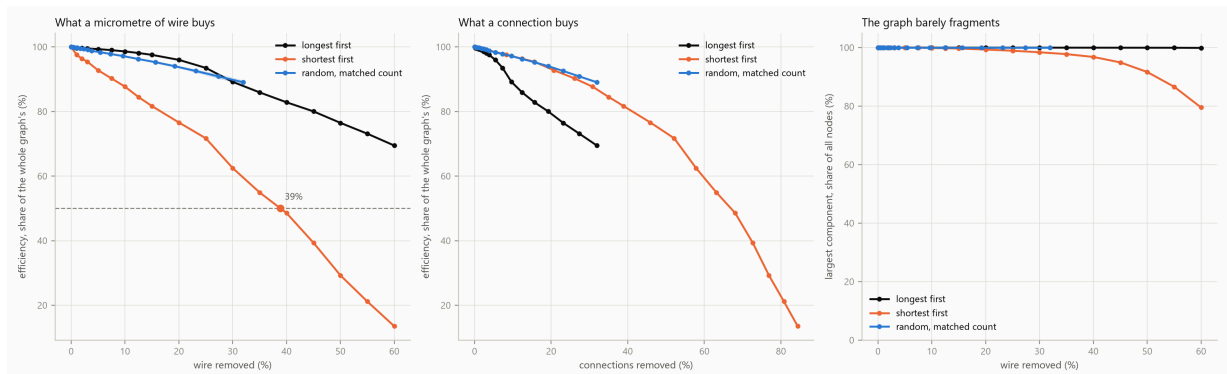


Figure 14: Efficiency against the wire removed and against the connections removed under three removal schedules, and the size of the largest weakly connected component.

- **Removal set.** Cutting only the neck crossings removes 963 units of flow; silencing every connection incident on a connective cell type removes 2,118. Both fall below their length-matched nulls overall and rise above them in the brain-to-nerve-cord direction (1.83 and 1.85).
- **Position measure.** Skeleton cable length tracks soma-to-output distance across classes ($\rho = 0.428$) but not within them ($|\rho| < 0.12$), so cost differences between classes are supported and differences between individual neurons of one class are not.
- **Swap search.** The greedy search was stopped before convergence, with the last 100,000 proposals still saving 0.33%, so the 33.64% reduction is a lower bound and not an estimate of the optimum.
- **Tail fit.** The power-law fit of the length distribution loses to both a lognormal and a truncated power law, and the longest connection is within 1% of the longest axis the somata span, so the tail is treated as bounded rather than scale-free.
- **Scope of the hub-placement correlations.** Degree against distance from the centre was taken over all types and over each compartment, with each scope measured from its own centroid. The two compartments do not agree in sign (+0.018 brain, -0.144 nerve cord), so they are reported apart: pooling them would cancel two weak tendencies into one number belonging to neither.
- **Null construction for side symmetry.** Of the two nulls, only the sign flip tests a direction. The random re-matching establishes that named counterparts are more alike than arbitrary neuropils and nothing more, so the size of a pair's difference is described rather than tested against an expectation.
- **Effect size against sample size.** The correlation between connection length and synapse count returns a p-value at the floor of double precision at $\rho = -0.054$. Over 490,884 connections that reflects the number of connections and not the size of the effect, and the correlation is read as the null it is.
- **Weighting of the region network.** Only the wire-weighted comparisons of the 89-region network carry information. At 68.9% density the unweighted clustering and path length are within 1% of their nulls, so the unweighted small-world question is degenerate rather than answered.
- **Matching for the removal schedules.** Matched on wire, the short end of the length distribution buys 4.4 times more efficiency per unit removed; matched on the number of connections, the longest connections are individually worth more than typical ones. Both matchings are reported, and the largest component distinguishes none of the three schedules.

Discussion

The fly's CNS is laid out economically in the weaker sense later work settled on, not in the strong sense of Cherniak's near-optimal layouts. At cell-type resolution, across brain, nerve cord and connective, the real layout is less than half as costly as random placements of the same positions. It is a quarter cheaper even when types are shuffled only within their compartment. The hemibrain neuron shuffles of Salova and Kovács (2025) found real layouts about half as costly as shuffled ones; the ratio here, 0.459, is similar at a different resolution and over the whole CNS.

Like the *C. elegans* layout, the fly's is far from the cheapest possible arrangement: greedy swaps remove at least a third of its wire, a margin close to Gushchin and Tang's for worm interneurons. Wiring cost is one pressure among several. The long connections that keep the layout from its optimum are the kind Kaiser and Hilgetag (2006) argued shorten processing paths.

The economy is also local. Attributed to the neuropils, the budget is dominated by a few regions, and inside every region large enough to test the cell types are placed more cheaply than permutations of their own positions. The clearest exceptions are the antennal lobes, whose internal wiring is within a few per cent of the permuted mean in both hemispheres, so whatever fixes the position of a cell type inside an antennal lobe, it is not the length of the connections between them.

How the budget is spread over connections sets the scale for what follows. It is uneven, with a Gini of 0.404 and nearly a third of the wire in the longest tenth of the connections, but the tail is not extreme: a lognormal describes it better than a power law does, and the longest connection spans about the longest axis the somata occupy. Nothing in the length distribution alone marks out a small set of connections as a class apart. What marks the neck-crossing connections out is where they run, not how unusual their length is.

The connective is where those long connections concentrate. It is 7.5% of connections and a quarter of the wire, and in two respects it looks like the hub backbone of mammalian connectomes. Its cell types are over-represented among the most connected types of the CNS, as Lin et al. (2024) predicted, and it favours well-connected partners on both sides. The backbone analogy is weaker than it first appears even here: high-degree cell types attract longer connections, but the 41.8% of connections incident on one carry only 46.4% of the cost.

The result the rich-club framing most directly predicts is that the connective preferentially joins hubs of the brain to hubs of the nerve cord. That result is weaker:

- the excess is 10%, and at the registered threshold comes entirely from ascending routes, but which direction carries it changes with the input threshold;
- it disappears when weaker connections are counted;
- a model with no knowledge of degree produces as many such routes.

Partner enrichment and hub membership are robust; rich-to-rich routing specifically is not.

The value result reverses the simplest version of the hypothesis. The pre-registered comparison asked whether the connective buys more flow than random wiring of the same length. It buys less, because the same length buys four times as many short connections elsewhere, and flow capacity counts disjoint paths.

Per connection the connective carries more flow than ordinary connections, and against the longest ordinary connections of the same total length it carries the same flow: on this measure it is valued as long wiring in general is valued. Its distinctive contribution is directional, 1.83 times the brain-to-nerve-cord flow of random wiring of its length and less than half of the reverse. Nor does spending more wire buy more flow for individual cell types: most can be removed without loss, and among those that matter, value follows the number of connections rather than their length.

The generative model is the plainest negative result here, and it is worth reading as one rather than as a footnote to the route count. A logistic model of connection presence on distance, its logarithm, a shared-compartment term and a full set of ordered cell-class pairs reaches a cross-validated AUC of 0.847, and its synthetic graphs reproduce 2 of the 13 properties compared; adding source and target degree raises the AUC to 0.899 and reproduces one. Both models draw every ordered pair independently, so the reciprocity and transitivity of the real graph are beyond them by construction. But they also miss total wiring cost, median connection length, the number of neck crossings and flow capacity, none of which independence rules out. Two readings fit. The first is that distance, compartment and cell class are the wrong generators, and that whatever actually fixes the wiring is absent from the model. The second is that the criterion is strict: a property counts as reproduced only when the real value falls inside the central 95% of 50 synthetic values, and that interval is narrow. Total wiring cost is the clearest case. Model G's graphs cost 92,666 mm against the real 93,095 mm, half a per cent low, and the property counts as not reproduced. The negative result is therefore about calibration as much as about mechanism. What it does settle is the one thing the protocol was registered to settle either way: the single property model G reproduces is the rich-to-rich route count, the statistic the rich-club section had read as evidence of structure.

The fly places its most expensive wiring on well-connected cell types, as the human connectome does. On these measures, however, that wiring does not return more sensory-to-motor capacity per unit length than ordinary long wiring, except in the brain-to-nerve-cord direction.

What this does and does not show

This analysis shows five things about one male fly CNS:

- the placement of its cell types is cheaper than random placement, and far from the cheapest arrangement reachable by swaps;
- the placement is economical inside the neuropils as well as across them, in every region large enough to test but the antennal lobes;
- the connective's cell types sit among its hubs, and its connections favour well-connected partners;
- how much the connective's wiring contributes to one measure of sensory-to-motor routing, measured against three explicitly constructed comparison sets;
- five further descriptions of the same wiring, added after the registered tests: where the best connected cell types sit, how the wire divides between the two sides of the body, what a micrometre of it buys in synapses, how the neuropils are arranged as a network, and what each end of the length distribution is worth to the graph.

Four of the thirteen registered hypotheses failed, and the failures are as much a part of the result as the successes. H4: cutting the neck costs less flow than random wiring of the same total length, not more, so the connective does not buy above-average sensory-to-motor capacity for what it spends. H10 and H10b: across individual connective cell types, wire length does not predict flow, and the correlation that exists among descending types is carried by the number of connections rather than by their length. H11: the rich-to-rich route excess does not hold at a 0.5% input threshold, and the direction it comes from changes across the sweep, so neither the excess nor its ascending origin is a threshold-independent property of the connective.

The five later analyses are descriptions of this specimen rather than tests of a hypothesis, and three of them return a null where a wiring-economy reading would have predicted something: degree does not predict how centrally a cell type sits, the two hemispheres differ no more than chance allows, and length does not predict synapse count. None of the five changes the thirteen registered hypotheses or the four that failed.

It does not show that wiring economy is a law of nervous-system organization, or that the placement is optimal. The generative model reproduces or fails to reproduce specific statistics; it does not describe how the nervous system develops. Flow capacity is a structural count of disjoint paths, not a measure of information transmitted, of behaviour, or of anything cognitive, and a cell type that costs no flow when removed from the graph has not been shown to be dispensable to an animal. The efficiency of the removal schedules is the same kind of measure, over shortest paths rather than disjoint ones, and the two communities of the region network are a grouping of anatomy by wire, not functional modules. The results extend the FlyWire brain analyses cited above to data those studies did not have; they do not revise them.

Limitations

- **Resolution.** The graph is aggregated to cell types by hemisphere, and each type is placed at the centroid of its somata. Within a class, soma-to-output distance does not predict a neuron's cable length. Positions therefore capture differences in wiring cost between classes, not the cable of individual neurons.
- **Cost measure.** Euclidean distance between centroids stands in for the length of real processes, which follow neuropil tracts rather than straight lines.
- **Neuropil assignment.** A cell type is attributed to the neuropil surface nearest its position, which is the centroid of its cell bodies rather than of its synapses. Cell bodies lie in a rind outside the neuropils, so the attribution follows where a type's somata sit, and the wire it charges to a region is a property of its endpoints, not of the tract its processes take.
- **Tail fit.** Connection length is bounded by the size of the animal, so the power-law comparison weighs imperfect descriptions of a truncated distribution against each other rather than testing a mechanism.
- **Input threshold.** Every headline result uses a 1% input threshold. The placement and hub results hold from 0.5% to 5%; the rich-to-rich route result does not hold at 0.5%, and its split between ascending and descending routes holds at no two thresholds alike.
- **Multiplicity.** Two families of tests are larger than the hypotheses that govern them: the seven richness thresholds, of which only the top 10% was registered, and the 48 within-neuropil placement tests, which were not registered at all. The three significant richness thresholds survive a Bonferroni correction across the seven, and the 41 neuropils at the smallest attainable p fall below $0.05/48$; the remaining thresholds and regions are uncorrected exploratory tests, including the two neuropils that straddle an uncorrected $\alpha = 0.05$.
- **The connective set.** It is defined by majority superclass. Sensory ascending neurons also pass through the neck, but they are treated as sensory inputs because their cell bodies lie outside the CNS. Their neck-crossing wiring is therefore not part of the removed set.
- **Value measure.** Flow capacity counts unweighted edge-disjoint paths, so it rewards many short connections over fewer long ones. For that reason the length-matched comparison is reported alongside count-matched and longest-connection comparisons, not replaced by them.
- **Comparability with Fault Lines.** The flow metric and its code are those of Fault Lines, but the sets are not. Descending neurons, counted there among motor outputs, are excluded here because they are part of what is removed, and the graph is resolved by hemisphere rather than pooled across sides. Intact flow capacity is 8,731 here against 10,647 there, so capacities and losses cannot be compared between the two studies.
- **Swap search.** The search is greedy and was stopped before convergence. It also ignores the physical constraints that fix where neuropils and cell-body layers can lie.
- **Generative models.** Both draw each ordered pair independently, so they cannot produce reciprocity or clustering, and the central-95% criterion for reproducing a property is strict enough that a model can be within a per cent of the real value and still fail it.
- **Exploratory additions.** The wire atlas, the concentration of the wiring budget, the within-neuropil placement tests and five further analyses — hub placement, the wire on the two sides, synapses against length, the neuropil network, and removal from each end of the length distribution — were added after the registered analyses were run. They carry no hypothesis, no direction fixed in advance and no correction for multiplicity, and are reported as description.
- **Sampled efficiency.** The removal curves estimate global efficiency from one fixed sample of 300 source cell types, used at every point of every schedule. The curves are therefore comparable with each other, but each point is an estimate of the whole-graph value rather than the value itself.

- **The region network.** It inherits the neuropil attribution above, so a region's wire follows where the cell bodies of its types sit. At 68.9% density its unweighted structure carries almost no information and only the wire-weighted comparisons say anything, and its two communities are a partition of anatomy by wire.
- **Side symmetry.** Neither null establishes how alike two copies of one structure should be. The sign flip can show a consistent bias and does not, and the random re-matching compares counterparts with unrelated neuropils, which is a much weaker standard.
- **Data.** The data are one animal, one sex and one static reconstruction.
- **Source version.** The flow analysis of Berg et al. (2026) was read in its bioRxiv version (v2, October 2025); the published version could differ.

References

- Ahn Y-Y, Jeong H, Kim BJ (2006). Wiring cost in the organization of a biological neuronal network. *Physica A* 367:531–537. doi:10.1016/j.physa.2005.12.013
- Alstott J, Bullmore E, Plenz D (2014). powerlaw: a Python package for analysis of heavy-tailed distributions. *PLoS ONE* 9(1):e85777. doi:10.1371/journal.pone.0085777
- Bates AS, Phelps JS, Kim M, Yang HH, et al. (2026). Distributed control circuits across a brain-and-cord connectome. *Nature* 656(8129):957–970. doi:10.1038/s41586-026-10735-w
- Berg S, et al. (2026). Sexual dimorphism in the complete *Drosophila* male central nervous system connectome. *Cell* 189(18):5504–5526.e15. doi:10.1016/j.cell.2026.08.015
- Chen BL, Hall DH, Chklovskii DB (2006). Wiring optimization can relate neuronal structure and function. *PNAS* 103(12):4723–4728. doi:10.1073/pnas.0506806103
- Cherniak C (1994). Component placement optimization in the brain. *J Neurosci* 14(4):2418–2427. doi:10.1523/JNEUROSCI.14-04-02418.1994
- Cherniak C (1995). Neural component placement. *Trends Neurosci* 18(12):522–527. doi:10.1016/0166-2236(95)98373-7
- Clauset A, Shalizi CR, Newman MEJ (2009). Power-law distributions in empirical data. *SIAM Review* 51(4):661–703. doi:10.1137/070710111
- Colizza V, Flammini A, Serrano MA, Vespignani A (2006). Detecting rich-club ordering in complex networks. *Nat Phys* 2(2):110–115. doi:10.1038/nphys209
- Dorkenwald S, Matsliah A, Sterling AR, et al. (2024). Neuronal wiring diagram of an adult brain. *Nature* 634:124–138. doi:10.1038/s41586-024-07558-y
- Ford LR, Fulkerson DR (1956). Maximal flow through a network. *Canadian Journal of Mathematics* 8:399–404. doi:10.4153/CJM-1956-045-5
- Grindrod P, Lambiotte R, Sahasrabudde R (2024). Modularity, hierarchical flows and symmetry of the *Drosophila* connectome. arXiv:2412.13202
- Gushchin A, Tang A (2015). Total wiring length minimization of *C. elegans* neural network: a constrained optimization approach. *PLOS ONE* 10(12):e0145029. doi:10.1371/journal.pone.0145029
- Kaiser M, Hilgetag CC (2006). Nonoptimal component placement, but short processing paths, due to long-distance projections in neural systems. *PLoS Comput Biol* 2(7):e95. doi:10.1371/journal.pcbi.0020095
- Lin A, et al. (2024). Network statistics of the whole-brain connectome of *Drosophila*. *Nature* 634(8032):153–165. doi:10.1038/s41586-024-07968-y
- Maslov S, Sneppen K (2002). Specificity and stability in topology of protein networks. *Science* 296(5569):910–913. doi:10.1126/science.1065103
- Menger K (1927). Zur allgemeinen Kurventheorie. *Fundamenta Mathematicae* 10:96–115. doi:10.4064/fm-10-1-96-115
- Pedregosa F, Varoquaux G, Gramfort A, et al. (2011). Scikit-learn: machine learning in Python. *Journal of Machine Learning Research* 12:2825–2830. arXiv:1201.0490
- Péntek B, Ercsey-Ravasz M (2025). The exponential distance rule-based network model predicts topology and reveals functionally relevant properties of the *Drosophila* projectome. *Netw Neurosci* 9(3):869–895. doi:10.1162/netn_a_00455

- Plaza SM, Clements J, Dolafi T, et al. (2022). neuPrint: an open access tool for EM connectomics. *Frontiers in Neuroinformatics* 16:896292. doi:10.3389/fninf.2022.896292
- Ramón y Cajal S (1909–1911). *Histologie du système nerveux de l'homme et des vertébrés*, trans. L. Azoulay. Paris: Maloine. English translation by N. Swanson and L. W. Swanson (1995), *Histology of the Nervous System of Man and Vertebrates*, Oxford University Press.
- Rivera-Alba M, et al. (2011). Wiring economy and volume exclusion determine neuronal placement in the *Drosophila* brain. *Curr Biol* 21(23):2000–2005. doi:10.1016/j.cub.2011.10.022
- Salova A, Kovács IA (2025). Combined topological and spatial constraints are required to capture the structure of neural connectomes. *Netw Neurosci* 9(1):181–206. doi:10.1162/netn_a_00428
- Schlegel P, Yin Y, Bates AS, et al. (2024). Whole-brain annotation and multi-connectome cell typing of *Drosophila*. *Nature* 634:139–152. doi:10.1038/s41586-024-07686-5
- Sulyok B, Balogh SG, Palla G (2026). Network geometry of the *Drosophila* brain. arXiv:2602.16417
- Takemura S, Hayworth KJ, Huang GB, et al. (2024). A connectome of the male *Drosophila* ventral nerve cord. *eLife* 13:RP97769. doi:10.7554/eLife.97769
- Towlson EK, Vértés PE, Ahnert SE, Schafer WR, Bullmore ET (2013). The rich club of the *C. elegans* neuronal connectome. *J Neurosci* 33(15):6380–6387. doi:10.1523/JNEUROSCI.3784-12.2013
- van den Heuvel MP, Sporns O (2011). Rich-club organization of the human connectome. *J Neurosci* 31(44):15775–15786. doi:10.1523/JNEUROSCI.3539-11.2011
- van den Heuvel MP, Kahn RS, Goñi J, Sporns O (2012). High-cost, high-capacity backbone for global brain communication. *PNAS* 109(28):11372–11377. doi:10.1073/pnas.1203593109
- Vuong QH (1989). Likelihood ratio tests for model selection and non-nested hypotheses. *Econometrica* 57(2):307–333. doi:10.2307/1912557